Advanced AC Circuit

Frequency Response

Words in Control System

Household appliances：

- heating and air-conditioning systems,
- switch-controlled thermostats,
- washers and dryers,
- cruise controllers in automobiles,
- elevators,
- traffic lights,
- manufacturing plants,
- navigation systems

In the aerospace field

In the manufacturing sector…

——Combination of circuit theory and communication theory

Outline

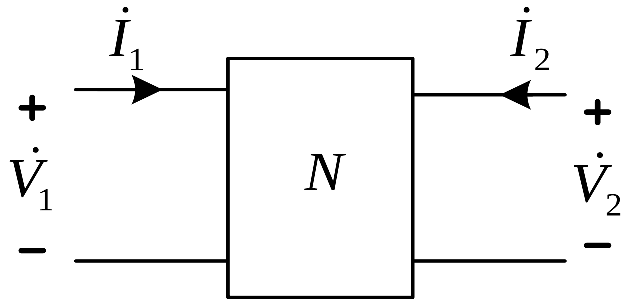
- Transfer Function and Frequency Response
- Filters
- Series/Parallel Resonance

Transfer Function (Network Function)

- **Transfer function**

— the frequency-dependent ratio of a phasor output *Y*(*ω*) to a phasor input *X*(*ω*).

$$H(\omega)=\frac{\dot{Y}(\omega)}{\dot{X}(\omega)}$$



- **Driving Point Function**

— the output and input are in the same port.

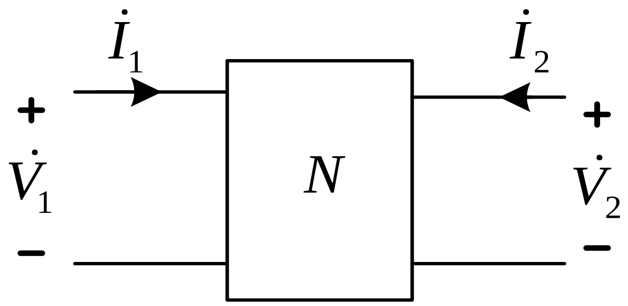
$$Z(\omega)=\frac{\dot{V}_1}{\dot{I}_1}$$ **— Driving point impedance**

$$H(\omega)=\frac{\dot{I}_1}{\dot{V}_1}$$ **— Driving point admittance**

Transfer Function (Network Function)

- **Transfer Function**

  — the output and input are in the different ports.



$$H(\omega)=\frac{\dot{V}_2}{\dot{V}_1}$$ **— Voltage Gain**

$$H(\omega)=\frac{\dot{I}_2}{\dot{I}_1}$$ **— Current Gain**

$$H(\omega)=\frac{\dot{V}_2}{\dot{I}_1}$$ **— Transfer impedance**

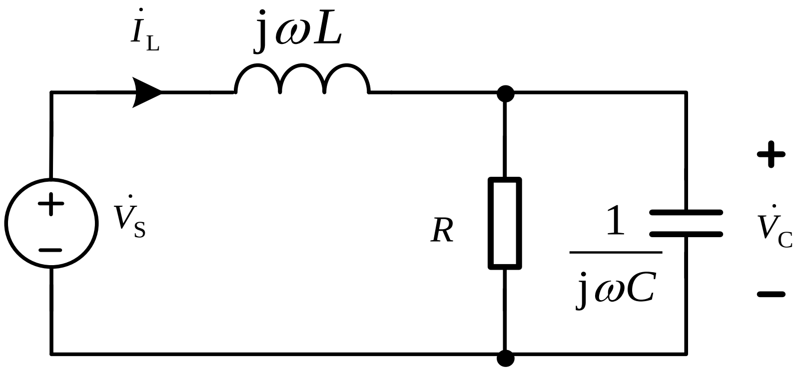
$$H(\omega)=\frac{\dot{I}_2}{\dot{V}_1}$$ **— Transfer admittance**

Transfer Function (Network Function)

Example **Obtain the transfer function V_C/V_S**

Solution

$$H(\omega)=\frac{\dot{V}_C}{\dot{V}_S}$$



$$H(\omega)=\frac{\dot{V}_C}{\dot{V}_S}=\frac{\dfrac{R/j\omega C}{R+1/j\omega C}}{j\omega L+\dfrac{R/j\omega C}{R+1/j\omega C}}$$

$$=\frac{R}{LRC(j\omega)^2+Lj\omega+R}=\frac{1/LC}{(j\omega)^2+j\omega/RC+1/LC}$$

Transfer Function is a ratio of two polynomials of $j\omega$

$H(\omega)$ can also be written as $H(j\omega)$.

Frequency Response

The frequency response of a circuit

—the variation in its behavior with change in signal frequency.

- **Magnitude Response**

$$a(\omega) = \left| H(\mathrm{j}\omega) \right| = Y_{\mathrm{m}}(\omega) / X_{\mathrm{m}}(\omega)$$

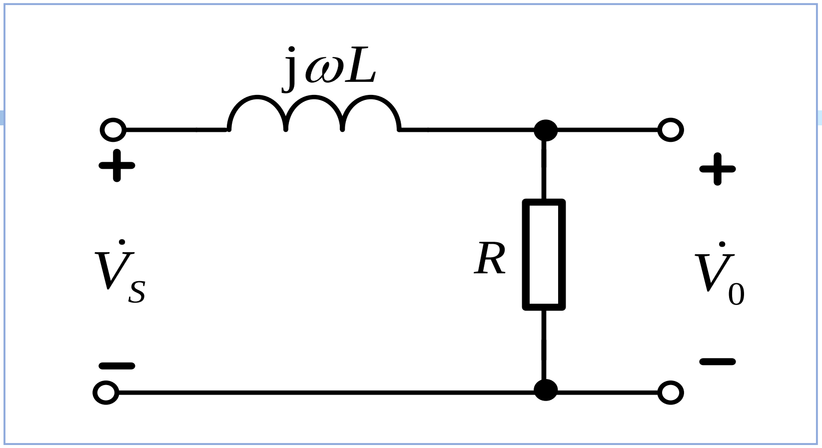
- **Phase Response**

$$\theta(\omega) = \angle H(\mathrm{j}\omega) = \phi_{\mathrm{y}} - \phi_{\mathrm{x}}$$

- **Magnitude Response Plot：** $a(\omega) \sim \omega$
- **Phase Response Plot：** $\theta(\omega) \sim \omega$

Frequency Response

Example $R = 100\Omega \quad L = 0.01\text{H}$

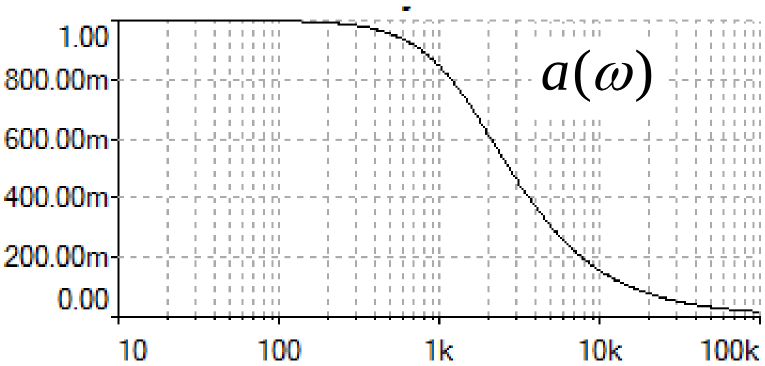


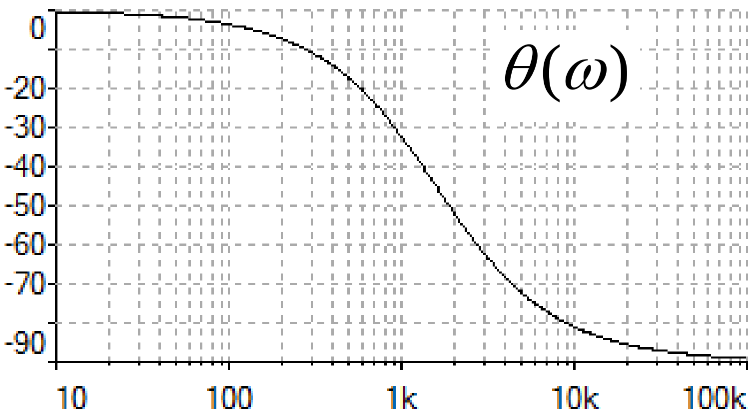
$$H(\mathrm{j}\omega) = \frac{\dot{V}_o}{\dot{V}_s} = \frac{R}{\mathrm{j}\omega L + R} = \frac{R/L}{\mathrm{j}\omega + R/L}$$

$$H(\mathrm{j}\omega) = \frac{10^4}{\mathrm{j}\omega + 10^4}$$

$$a(\omega) = |H(\mathrm{j}\omega)| = \frac{1}{\sqrt{1+\omega^2/10^8}}$$

$$\theta(\omega) = \angle H(\mathrm{j}\omega) = -\tan^{-1}\frac{\omega}{10^4}$$





The Decibel Scale

- **Bel — measurement for Power Gain**

$$G_{\text{bel}} = \log_{10}\frac{P_2}{P_1}$$

- **Decibel — 1/10th of a bel**

$$G_{\text{dB}} = 10\log_{10}\frac{P_2}{P_1}$$

When P_2=P_1 $G_{\text{dB}} = 10\lg\frac{P_2}{P_1} = 10\lg 1 = 0\text{dB}$

When P_2=2P_1 $G_{\text{dB}} = 10\lg\frac{P_2}{P_1} = 10\lg 2 = 3\text{dB}$

When P_2=0.5P_1 $G_{\text{dB}} = 10\lg\frac{P_2}{P_1} = 10\lg 0.5 = -3\text{dB}$

The Decibel Scale

- **Decibel —— comparing for levels of voltage or current (on one resistor)**

$$G_{\mathrm{dB}}=10\log_{10}\frac{P_2}{P_1}=10\lg\frac{V_{2\mathrm{m}}{}^2/2R}{V_{1\mathrm{m}}{}^2/2R}=20\lg\frac{V_{2\mathrm{m}}}{V_{1\mathrm{m}}}$$

$$G_{\mathrm{dB}}=10\log_{10}\frac{P_2}{P_1}=10\lg\frac{I_{2\mathrm{m}}{}^2R/2}{I_{1\mathrm{m}}{}^2R/2}=20\lg\frac{I_{2\mathrm{m}}}{I_{1\mathrm{m}}}$$

When P_2=P_1 $\quad G_{\mathrm{dB}}=10\lg\frac{P_2}{P_1}=10\lg 1=0\mathrm{dB} \qquad V_{2\mathrm{m}}=V_{1\mathrm{m}}$

When P_2=2P_1 $\quad G_{\mathrm{dB}}=10\lg\frac{P_2}{P_1}=10\lg 2=3\mathrm{dB} \qquad V_{2\mathrm{m}}=\sqrt{2}V_{1\mathrm{m}}$

When P_2=0.5P_1 $\quad G_{\mathrm{dB}}=10\lg\frac{P_2}{P_1}=10\lg 0.5=-3\mathrm{dB} \qquad V_{2\mathrm{m}}=V_{1\mathrm{m}}/\sqrt{2}$

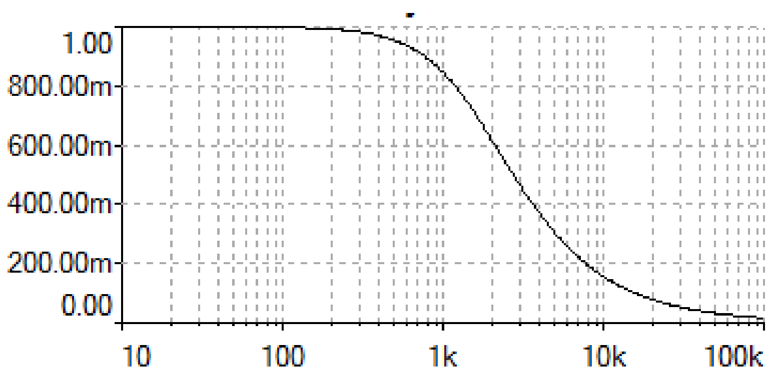
Bode Plots

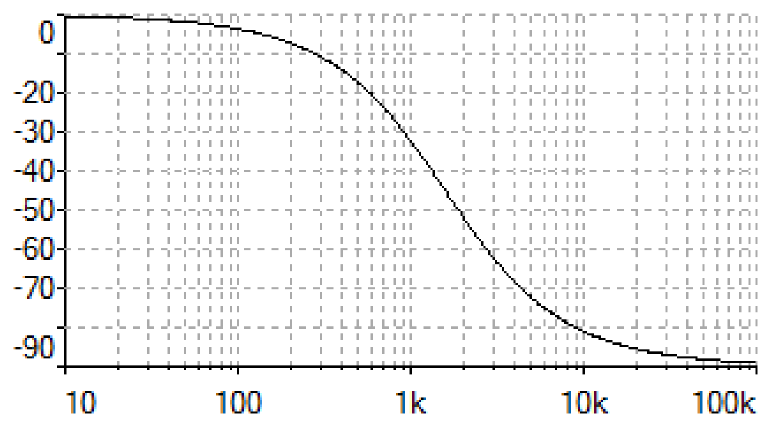
—Semi-logarithm plots of the magnitude (in ratio value) and phase (in degrees) of a transfer function versus frequency.

Magnitude axis — linear scale

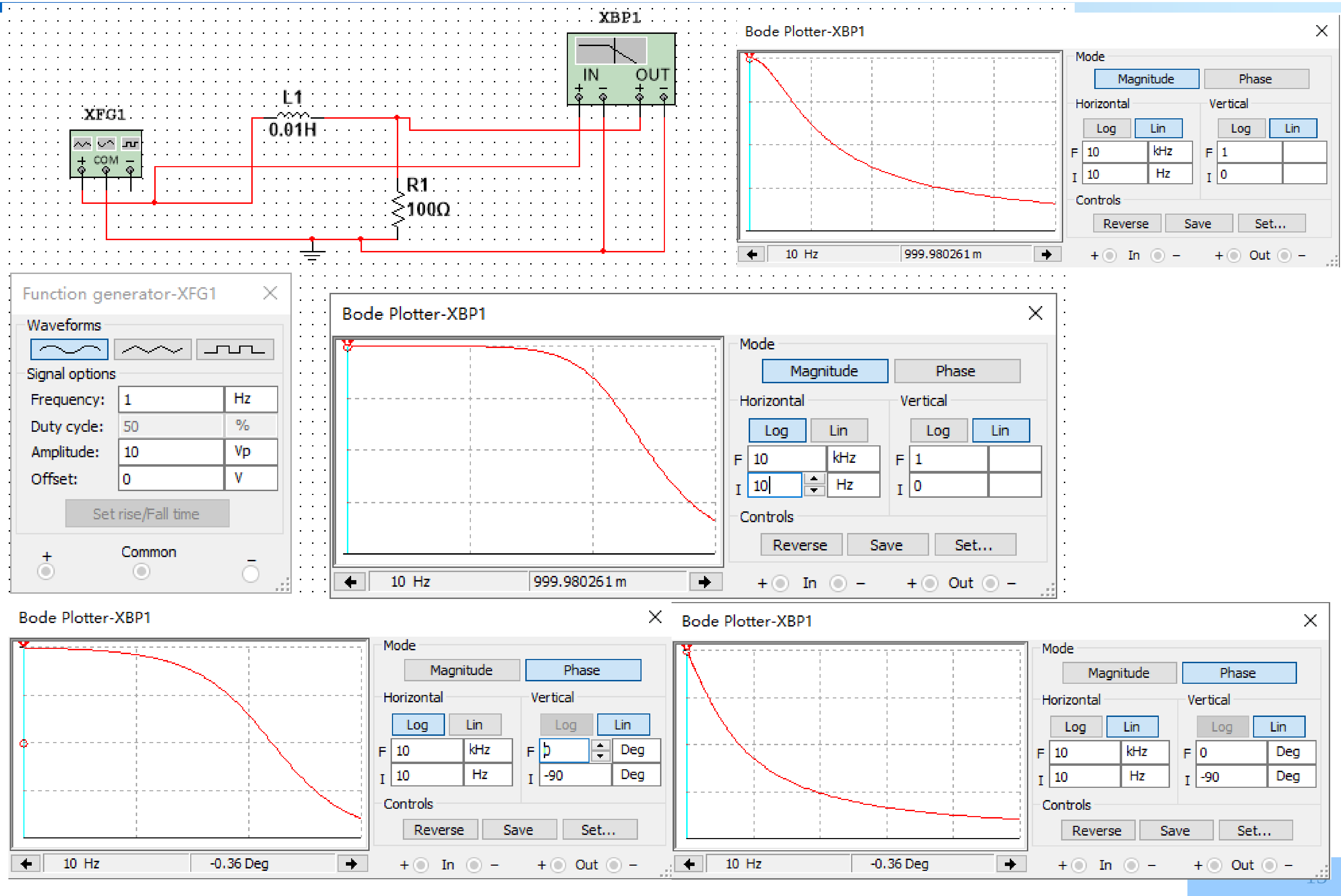
Phase axis — linear scale

Frequency axis — logarithmic scale





Bode Plot Simulation



Homework

Problem 1, 2, 3

- **Transfer Function and Frequency Response**
- **Filters**
- **Series/Parallel Resonance**

Filters

- **Filter**

  —— a circuit that is designed to pass signals with desired frequencies and reject or attenuate others.

- **Passive Filter**

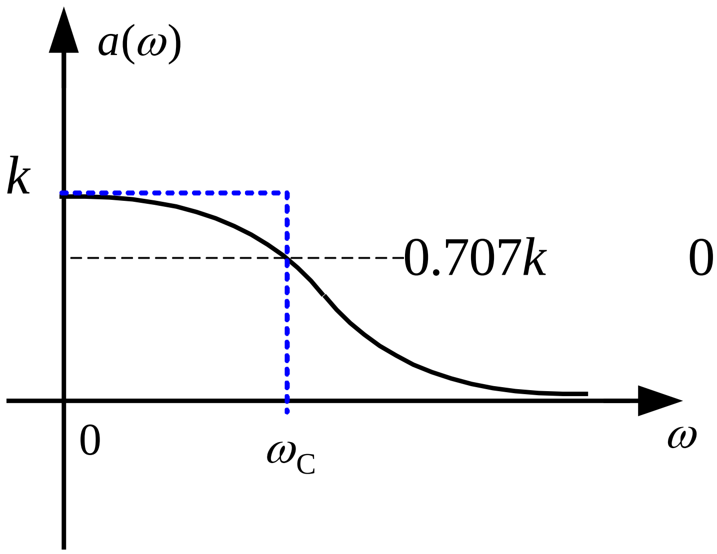
  —— a filter consists of only passive elements R, L and C.

- **Active Filter**

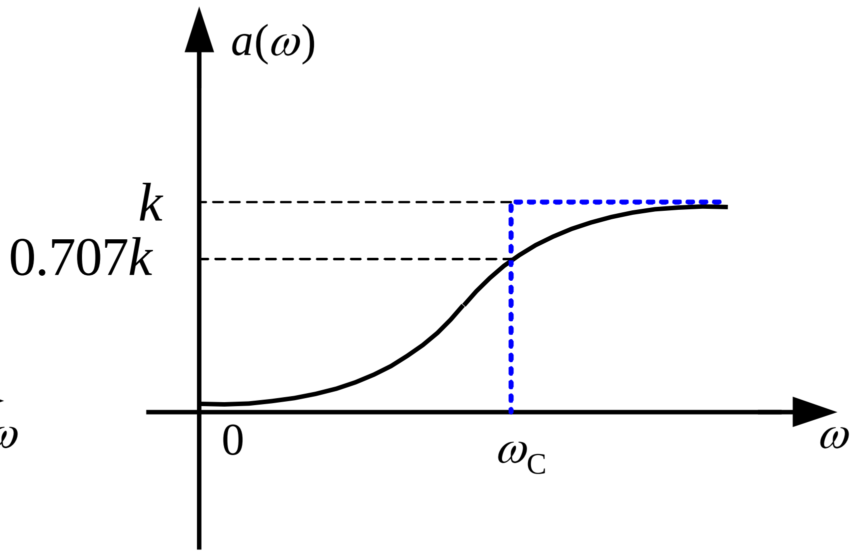
  —— a filter consists of active elements such as transistors and op amps in addition to passive elements R, L and C.

Types of Filters

Four types of filters according to magnitude response:



Lowpass Filter

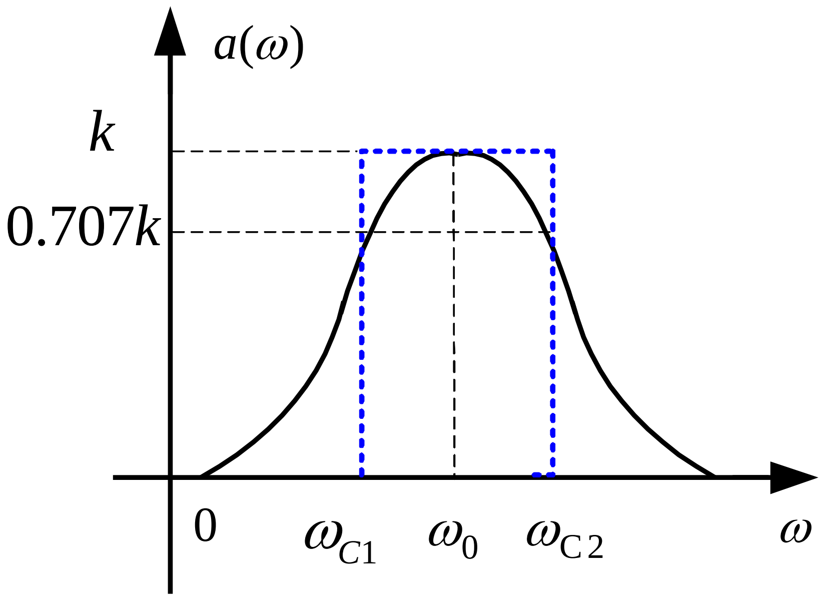


Highpass Filter

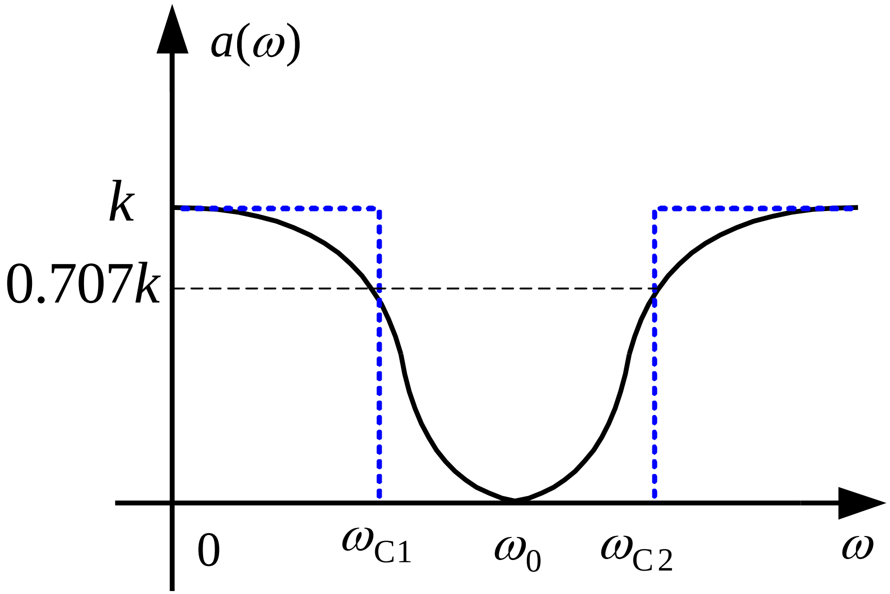
- **Cutoff frequency $f_C(\omega_C)$： the frequency at which the transfer function H drops in magnitude to 70.71% of its maximum value.**

Types of Filters

- **Center frequency $f_0(\omega_0)$**
- **Rejection frequency $f_0(\omega_0)$**



Bandpass Filter

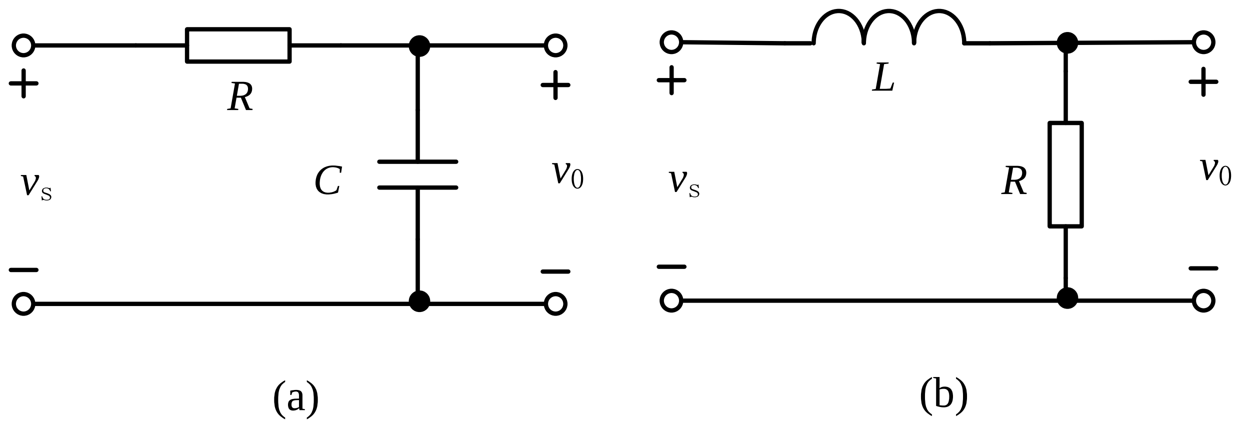


Bandstop Filter

k: Gain in band

The First Order Lowpass

- **Example**



（a） $$H(j\omega)=\frac{\dot{V}_o}{\dot{V}_s}=\frac{1/j\omega C}{1/j\omega C+R}=\frac{1/RC}{j\omega+1/CR}=\frac{1}{1+j\omega CR} \qquad \omega_c=\frac{1}{RC}$$

（b） $$H(j\omega)=\frac{\dot{V}_o}{\dot{V}_s}=\frac{R}{j\omega L+R}=\frac{R/L}{j\omega+R/L}=\frac{1}{1+j\omega L/R} \qquad \omega_c=\frac{R}{L}$$

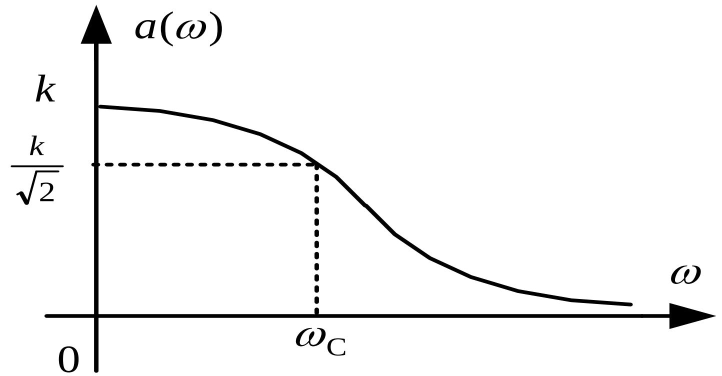
$$k=1$$

$$H(j\omega)=\frac{1}{1+j\omega/\omega_c}$$

The First Order Lowpass

- **General network Function for 1st order lowpass filter**

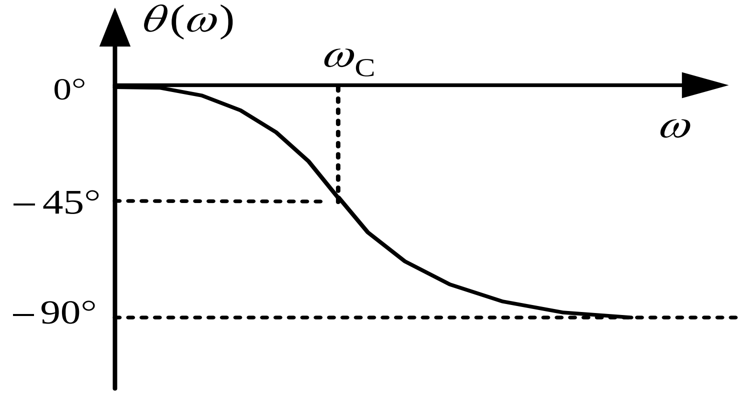
$$H(\mathrm{j}\omega)=\frac{k\omega_c}{\mathrm{j}\omega+\omega_c}=\frac{k}{1+\mathrm{j}\dfrac{\omega}{\omega_c}}$$



- **Frequency response**

$$a(\omega)=\frac{|k|}{\sqrt{1+(\dfrac{\omega}{\omega_c})^2}}$$

$$\theta(\omega)=-\tan^{-1}\frac{\omega}{\omega_c}$$

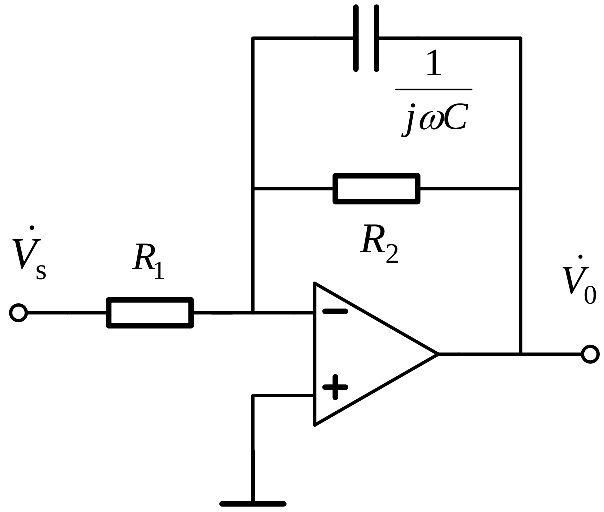


The First Order Lowpass

- **Example:** Demonstrate the circuit can be used as a lowpass filter. Select proper element value so that the band gain is 4 and cutoff angular frequency is 100rad/s.

Solution **Applying KCL at inverse input terminal, get**

$$\frac{\dot{V}_s}{R_1} = -\dot{V}_o\left(\frac{1}{R_2} + \mathrm{j}\omega C\right)$$



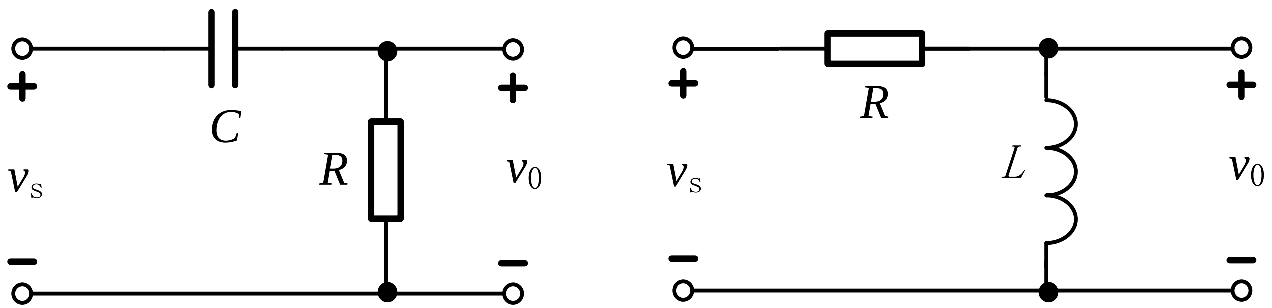
$$H(\omega) = \frac{\dot{V}_o}{\dot{V}_s} = -\frac{R_2}{R_1}\cdot\frac{1}{1+\mathrm{j}\omega C R_2} = k\cdot\frac{1}{1+\mathrm{j}\dfrac{\omega}{\omega_c}}$$

$$k = -\frac{R_2}{R_1} \qquad \omega_c = \frac{1}{R_2 C} \qquad |k| = \frac{R_2}{R_1} = 4$$

Select $R_1 = 10\mathrm{k}\Omega$ $\quad R_2 = 40\mathrm{k}\Omega$ $\quad C = \dfrac{1}{R_2\omega_c} = 250\,\mathrm{nF}$

The First Order Highpass

- **Example**



(a) (b)

（a） $$H(\omega)=\frac{R}{R+\frac{1}{\mathrm{j}\omega C}}=\frac{\mathrm{j}\omega}{\mathrm{j}\omega+1/RC}=\frac{\mathrm{j}\omega CR}{1+\mathrm{j}\omega CR}\qquad \omega_c=\frac{1}{RC}$$

（b） $$H(\omega)=\frac{\mathrm{j}\omega L}{R+\mathrm{j}\omega L}=\frac{\mathrm{j}\omega}{\mathrm{j}\omega+R/L}=\frac{\mathrm{j}\omega L/R}{1+\mathrm{j}\omega L/R}\qquad \omega_c=\frac{R}{L}$$

$$k=1$$

$$H(\mathrm{j}\omega)=\frac{\mathrm{j}\frac{\omega}{\omega_c}}{1+\mathrm{j}\frac{\omega}{\omega_c}}$$

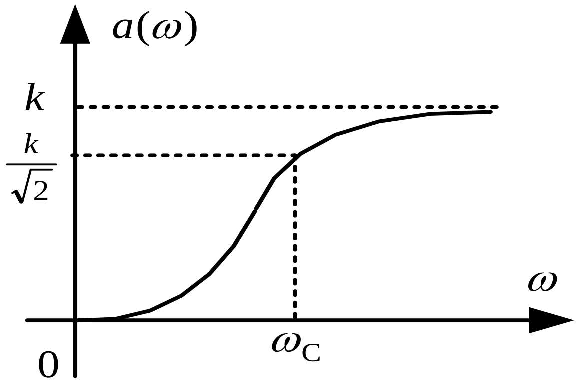
The First Order Highpass

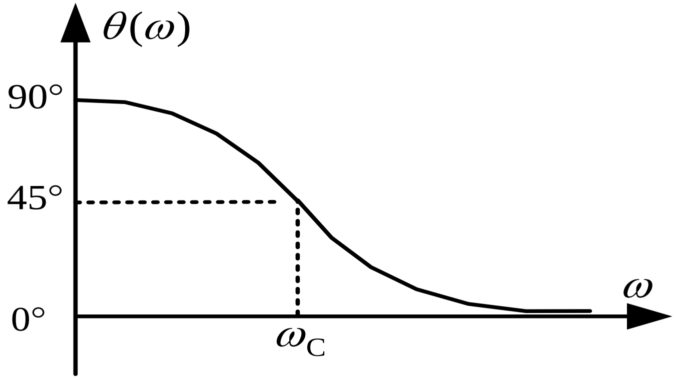
- **General network Function for 1st order highpass filter**

$$H(\mathrm{j}\omega) = k\frac{\mathrm{j}\omega}{\mathrm{j}\omega + \omega_\mathrm{c}} = k\frac{\mathrm{j}\dfrac{\omega}{\omega_\mathrm{c}}}{1 + \mathrm{j}\dfrac{\omega}{\omega_\mathrm{c}}}$$

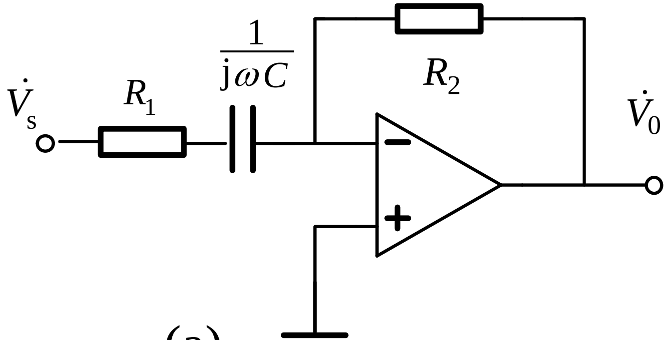
$$a(\omega) = \frac{k}{\sqrt{1 + \left(\dfrac{\omega_\mathrm{c}}{\omega}\right)^2}}$$

$$\theta(\omega) = \tan^{-1}\frac{\omega_\mathrm{c}}{\omega}$$

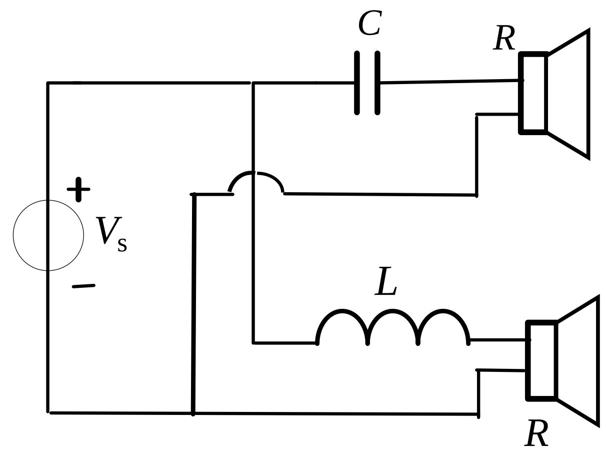




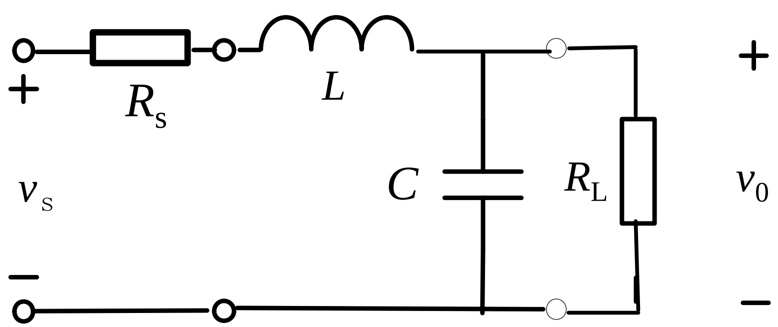
Exercises： Analyze the frequency response.



(a)



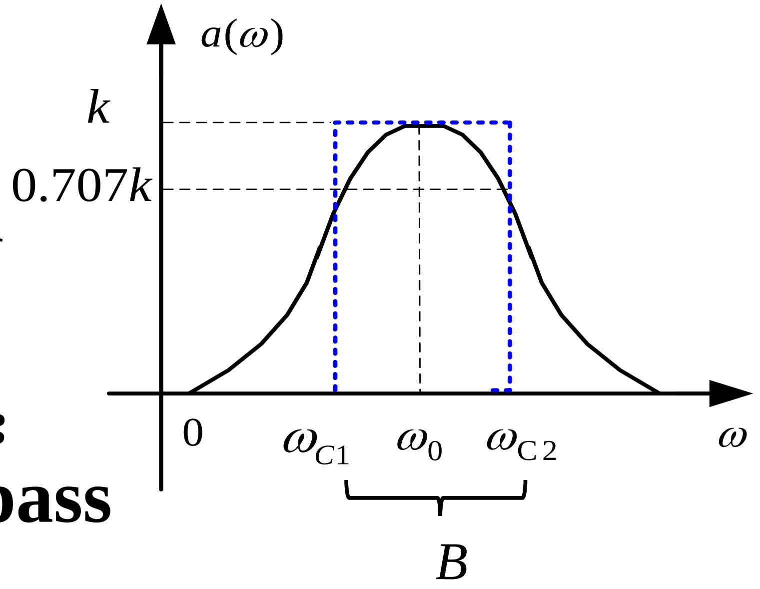
(b) Crossover network



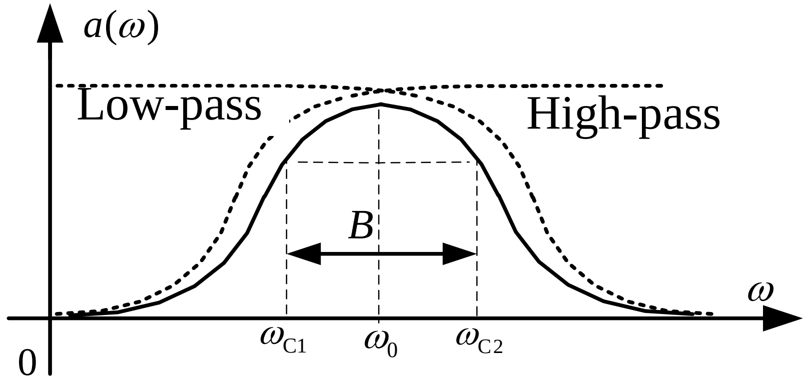
(c) Butterworth filter for $a(\omega)=\dfrac{k}{\sqrt{1+(\omega/\omega_c)^{2N}}}$

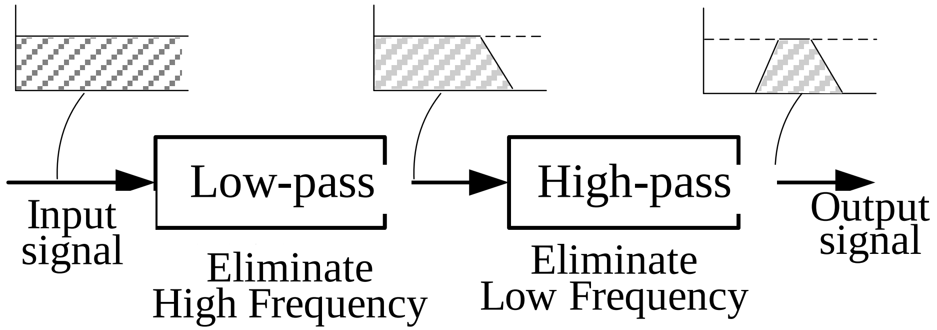
The 2nd Order Band-pass

- **A band-pass filter is designed to pass all frequencies within a band of $\omega_{c1} < \omega < \omega_{c2}$**



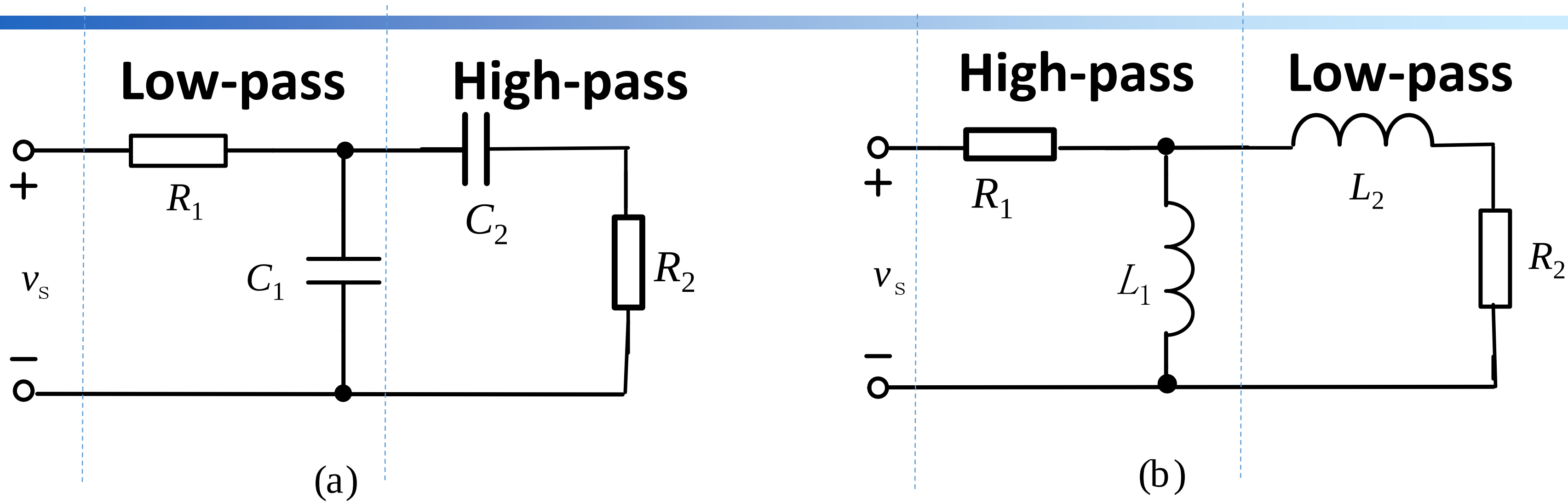
- **A method to obtain a band-pass: cascading a low-pass and a high-pass**





- **NOTE:** $f_{\text{c low-pass}} > f_{\text{c high-pass}}$

The 2nd Order Band-pass



(a) $$\dot{V}_o = \dot{V}_2 = \frac{Z_{C_1} // (Z_{C_2} + R_2)}{R_1 + Z_{C_1} // (Z_{C_2} + R_2)} \frac{R_2}{(Z_{C_2} + R_2)} \dot{V}_s$$

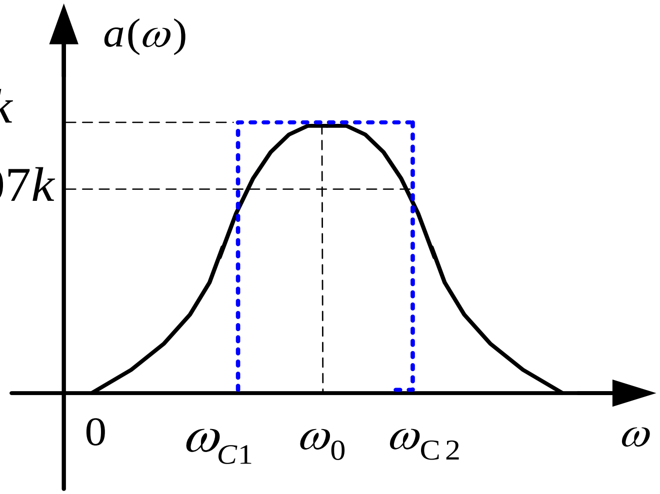
$$H_C = \frac{\dot{V}_2}{\dot{V}_s} = \frac{j\omega / C_1 R_1}{(j\omega)^2 + j\omega\left(\frac{1}{C_1 R_2} + \frac{1}{C_1 C_2} + \frac{1}{C_2 R_2}\right) + \frac{1}{C_1 R_1 C_2 R_2}}$$

(b) $$H_L = \frac{\dot{V}_2}{\dot{V}_s} = \frac{j\omega R_2 / L_2}{(j\omega)^2 + j\omega\left(\frac{R_1}{L_2} + \frac{R_1}{L_1} + \frac{R_2}{L_2}\right) + \frac{R_1 R_2}{L_1 L_2}}$$

The 2nd Order Band-pass

- **General network Function for 2nd order band-pass filter**

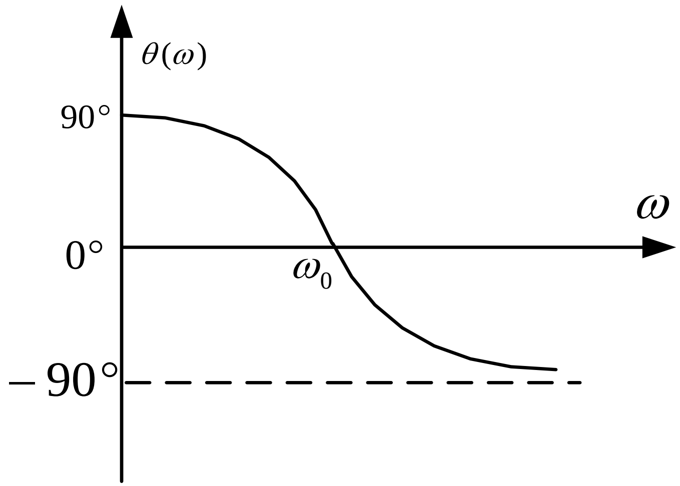
$$H(\mathrm{j}\omega)=k\frac{\frac{\omega_0}{Q}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\frac{\omega_0}{Q}\mathrm{j}\omega+\omega_0^2}$$



***k*: Passband Gain** **ω_0: Center Frequency, *Q*: Quality Factor**

$$a(\omega)=\frac{k}{\sqrt{1+Q^2\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)^2}}$$

$$\theta(\omega)=-\tan^{-1}Q\left(\frac{\omega}{\omega_0}-\frac{\omega_0}{\omega}\right)$$



The 2nd Order Bandpass

- **Quality Factor** $Q = \dfrac{\omega_0}{B}$
- **Bandwidth** $B = \omega_{c2} - \omega_{c1} = \dfrac{\omega_0}{Q}$

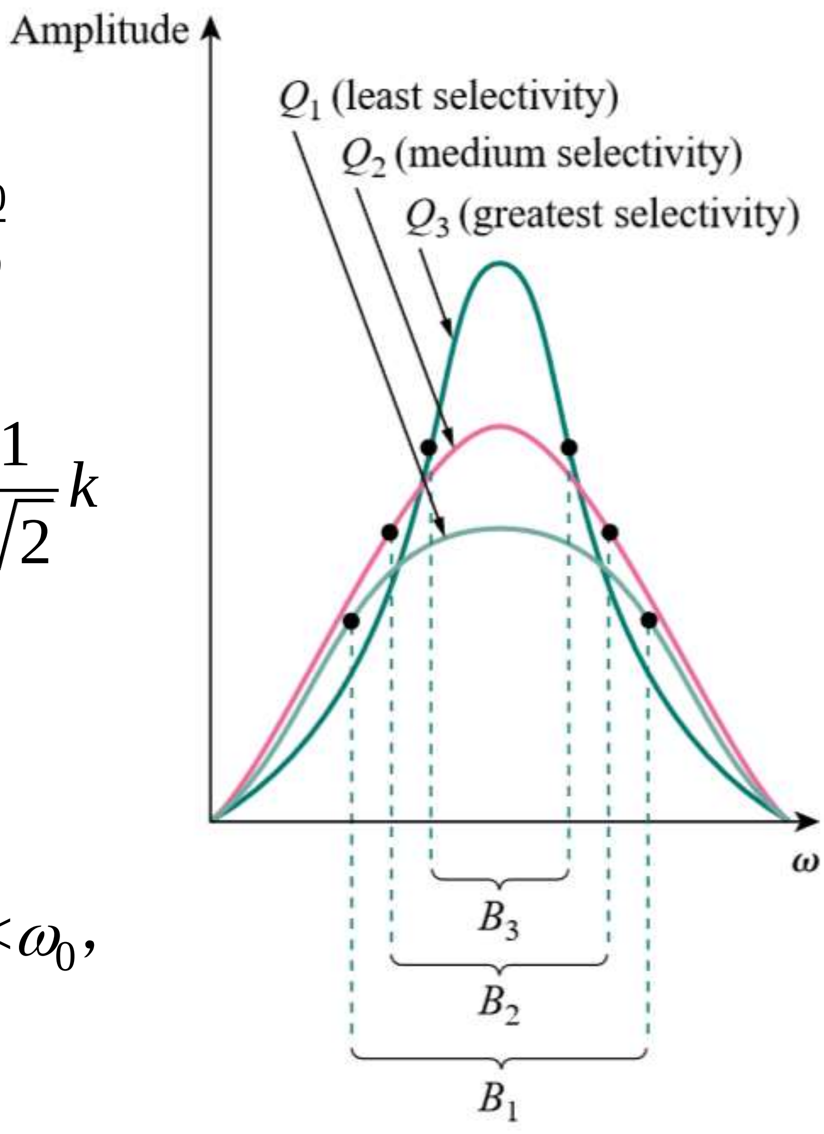
$$\omega_{c1} \cdot \omega_{c2} = \omega_0^2$$

Set $a(\omega_c) = \dfrac{k}{\sqrt{1+Q^2(\dfrac{\omega_c}{\omega_0}-\dfrac{\omega_0}{\omega_c})^2}} = \dfrac{1}{\sqrt{2}}k$

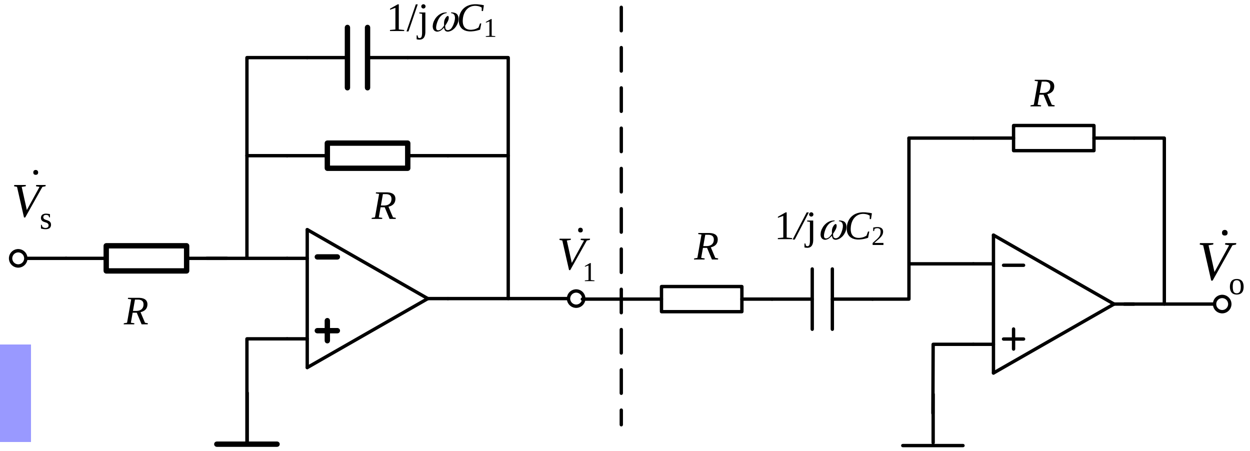
get $\omega_{c1}, \omega_{c2} = \dfrac{\omega_0}{2Q}(\sqrt{1+4Q^2} \mp 1)$

For 'high Q'circuit: $Q > 10$, $B \ll \omega_0$,

$$\omega_{c1}, \omega_{c2} \approx \omega_0 \mp \frac{B}{2}$$



Example Given C_1=100nF, C_2=400nF, R=1kΩ. Find $H(j\omega)$, ω_0, B, Q and ω_{C1}, ω_{C2}.



solution

First stage transfer function

$$H_1(\mathrm{j}\omega)=\frac{\dot{V}_1}{\dot{V}_s}=-\frac{1}{1+\mathrm{j}\omega C_1 R}=-\frac{1/RC_1}{\mathrm{j}\omega+1/RC_1}$$

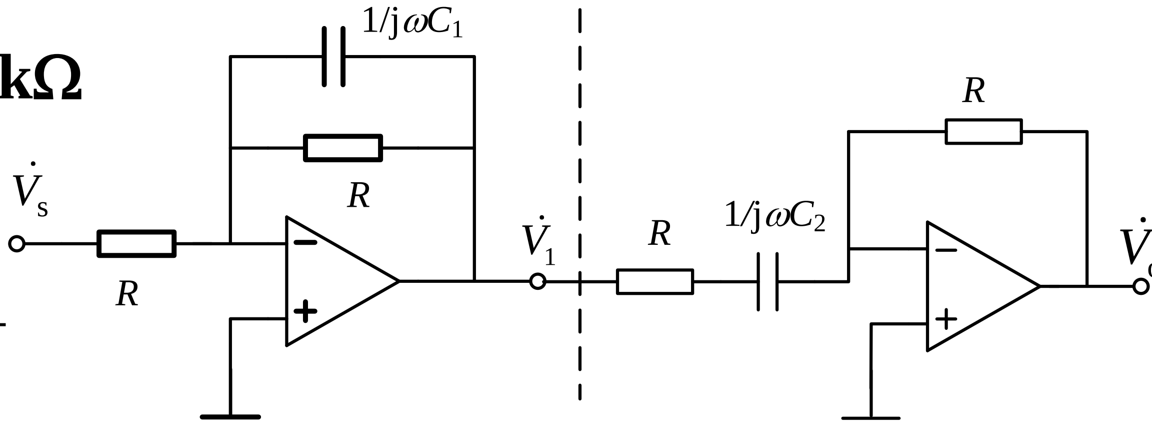
Second stage transfer function

$$H_2(\omega)=\frac{\dot{V}_o}{\dot{V}_1}=-\frac{\mathrm{j}\omega C_2 R}{1+\mathrm{j}\omega C_2 R}=-\frac{\mathrm{j}\omega}{\mathrm{j}\omega+1/RC_2}$$

Thus

$$H(\mathrm{j}\omega)=\frac{\dot{V}_o}{\dot{V}_S}=\frac{\dot{V}_o}{\dot{V}_1}\frac{\dot{V}_1}{\dot{V}_S}=H_1(\omega)H_2(\omega)=\frac{\mathrm{j}\omega/RC_1}{(\mathrm{j}\omega+1/RC_1)(\mathrm{j}\omega+1/RC_2)}$$

C_1=100nF, C_2=400nF, R=1kΩ



$$H(\mathrm{j}\omega)=k\frac{\dfrac{\omega_0}{Q}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\dfrac{\omega_0}{Q}\mathrm{j}\omega+\omega_0^2}$$

$$H(\mathrm{j}\omega)=\frac{\mathrm{j}\omega\cdot 10^4}{(\mathrm{j}\omega+10^4)(\mathrm{j}\omega+2500)}=\frac{\mathrm{j}\omega\cdot 10^4}{(\mathrm{j}\omega)^2+12500\cdot\mathrm{j}\omega+25\times 10^6}$$

$$\omega_0=5000\ \mathrm{rad/s}\quad \begin{cases} B=\omega_{c2}-\omega_{c1}=\dfrac{\omega_0}{Q}=12500\ \mathrm{rad/s} \\ \omega_{c1}\cdot\omega_{c2}=\omega_0^2 \end{cases}\qquad Q=\frac{\omega_0}{12500}=0.4$$

$$k=0.8$$

$$\omega_{c1}=\frac{\omega_0}{2Q}(\sqrt{1+4Q^2}-1)=\frac{\omega_0}{Q}(0.14)=1754\ \mathrm{rad/s}$$

$$\omega_{c2}=\frac{\omega_0}{2Q}(\sqrt{1+4Q^2}+1)=\frac{\omega_0}{Q}(1.14)=14254\ \mathrm{rad/s}$$

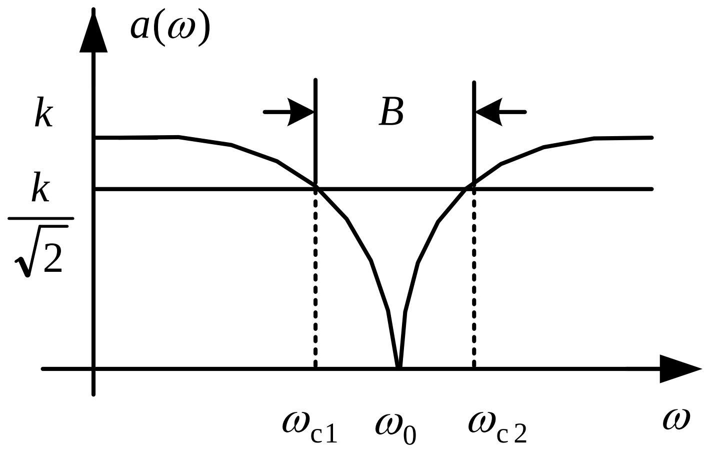
The 2nd Order Bandstop(Bandreject)*

- **General network Function for 2nd order bandstop filter**

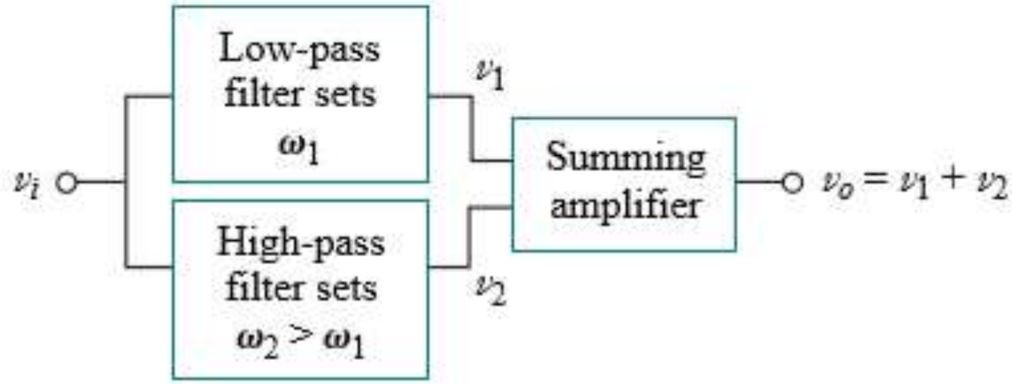
$$H(\mathrm{j}\omega)=k\frac{(\mathrm{j}\omega)^2+2\beta\cdot\mathrm{j}\omega+\omega_0^2}{(\mathrm{j}\omega)^2+\dfrac{\omega_0}{Q}\cdot\mathrm{j}\omega+\omega_0^2}\qquad \beta<<\frac{\omega_0}{2Q}$$

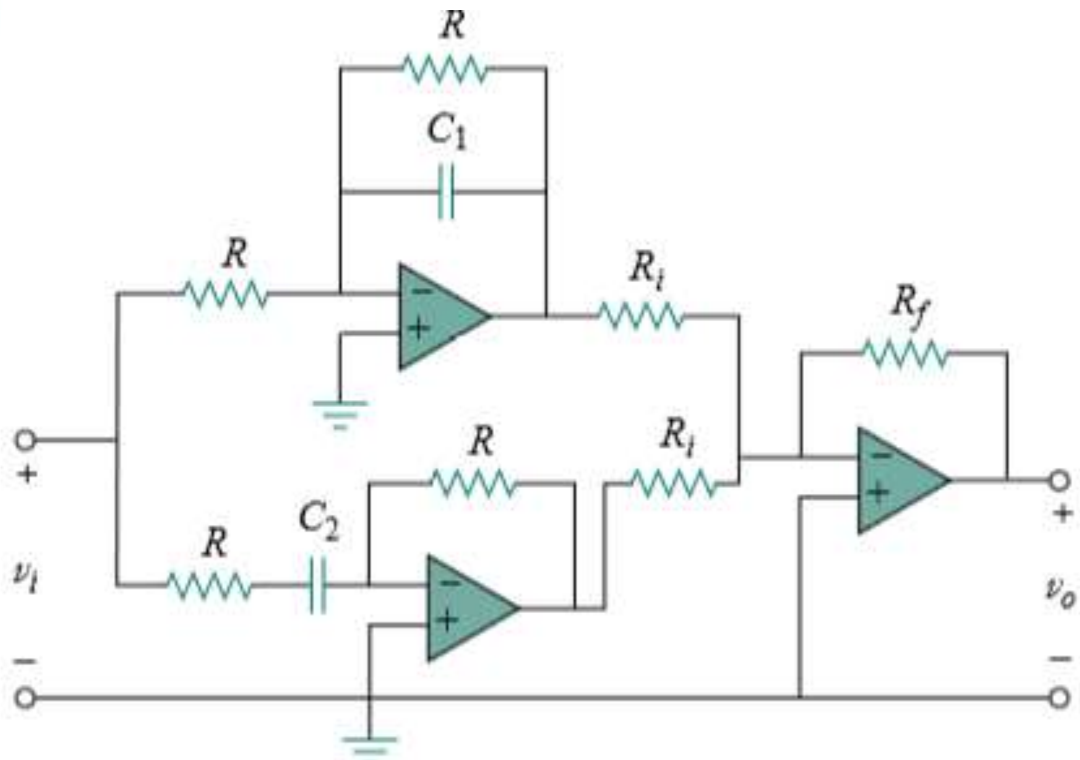
- **Amplitude frequency response**

Stopband width $B=\dfrac{\omega_0}{Q}$



A Method to Obtain a Band-stop：Paralleling a Low-pass and a High-pass





$$H(\mathrm{j}\omega) = \frac{R_\mathrm{f}}{R_\mathrm{i}} \frac{(\mathrm{j}\omega)^2 + 2\mathrm{j}\omega\omega_1 + \omega_1\omega_2}{(\mathrm{j}\omega)^2 + \mathrm{j}\omega(\omega_1 + \omega_2) + \omega_1\omega_2}$$

Homework

- **5, 6, 7, 9, 11,14**

Recap

- **Transfer function** **— the frequency-dependent ratio of a phasor output (an element voltage or current) *Y*(*ω*) to a phasor input (source voltage or current) *X*(*ω*).**

$$H(\omega)=\frac{\dot{Y}(\omega)}{\dot{X}(\omega)}$$

$$H(\omega)=\text{voltage gain}=\frac{V_\text{o}(\omega)}{V_\text{i}(\omega)}$$

$$H(\omega)=\text{current gain}=\frac{I_\text{o}(\omega)}{I_\text{i}(\omega)}$$

$$H(\omega)=\text{transfer impedance}=\frac{V_\text{o}(\omega)}{I_\text{i}(\omega)}$$

$$H(\omega)=\text{transfer admittance}=\frac{I_\text{o}(\omega)}{V_\text{i}(\omega)}$$

Recap — Filter

- **1st order low-pass filter**

$$H(\mathrm{j}\omega)=\frac{k\omega_c}{\mathrm{j}\omega+\omega_c}=\frac{k}{1+\mathrm{j}\dfrac{\omega}{\omega_c}}$$

- **1st order high-pass filter**

$$H(\mathrm{j}\omega)=k\frac{\mathrm{j}\omega}{\mathrm{j}\omega+\omega_c}=k\frac{\mathrm{j}\dfrac{\omega}{\omega_c}}{1+\mathrm{j}\dfrac{\omega}{\omega_c}}$$

- **2nd order low-pass filter**

$$H(\mathrm{j}\omega)=\frac{k}{(\mathrm{j}\omega)^2+\dfrac{\omega_0}{Q}\mathrm{j}\omega+\omega_0^2}$$

- **2nd order high-pass filter**

$$H(\mathrm{j}\omega)=\frac{k(\mathrm{j}\omega)^2}{(\mathrm{j}\omega)^2+\dfrac{\omega_0}{Q}\mathrm{j}\omega+\omega_0^2}$$

- **2nd order band-pass filter**

$$H(\mathrm{j}\omega)=k\frac{\dfrac{\omega_0}{Q}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\dfrac{\omega_0}{Q}\mathrm{j}\omega+\omega_0^2}$$

- **2nd order band-stop filter**

$$H(\mathrm{j}\omega)=k\frac{(\mathrm{j}\omega)^2+2\beta\cdot\mathrm{j}\omega+\omega_0^2}{(\mathrm{j}\omega)^2+\dfrac{\omega_0}{Q}\cdot\mathrm{j}\omega+\omega_0^2}$$

$$\beta<<\frac{\omega_0}{2Q}$$

Recap —— Filter

Steps to judge the type of a filter: **Find H(jω) or |H(ω)|**

- $\omega=0,\ \ |H(0)|=\alpha(0)=|k|$
- $\omega\to\infty,\ \ |H(\infty)|=\alpha(\infty)=0$

Low-pass

- $\omega=0,\ \ |H(0)|=\alpha(0)=0$
- $\omega\to\infty,\ \ |H(\infty)|=\alpha(\infty)=|k|$

High-pass

- $\omega=0,\ \ |H(0)|=\alpha(0)=0$
- $\omega\to\infty,\ \ |H(\infty)|=\alpha(\infty)=0$

Band-pass

- $\omega=0,\ \ |H(0)|=\alpha(0)=|k|$
- $\omega\to\infty,\ \ |H(\infty)|=\alpha(\infty)=|k|$

Band-stop

Recap —— Filter

- **Cutoff frequency $f_C(\omega_C)$: the frequency at which the transfer function H drops in magnitude to 70.71% of its maximum value.**

—— half-power frequencies

$$|H(\omega_c)| = \frac{1}{\sqrt{2}}|H(0)|$$

$$\boxed{|H(\omega_c)|_{\text{dB}} - |H(0)|_{\text{dB}} = -3\text{dB}}$$

Low-pass

$$|H(\omega_c)| = \frac{1}{\sqrt{2}}|H(\infty)|$$

$$\boxed{|H(\omega_c)|_{\text{dB}} - |H(\infty)|_{\text{dB}} = -3\text{dB}}$$

High-pass

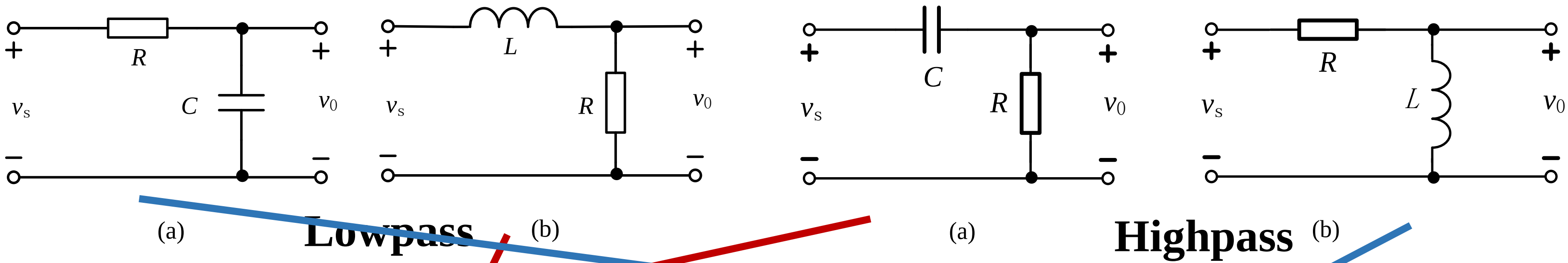
$$|H(\omega_c)| = \frac{1}{\sqrt{2}}|H(\omega_0)|$$

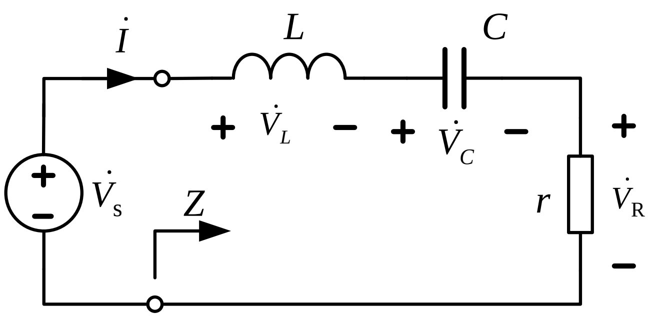
$$\boxed{|H(\omega_c)|_{\text{dB}} - |H(\omega_0)|_{\text{dB}} = -3\text{dB}}$$

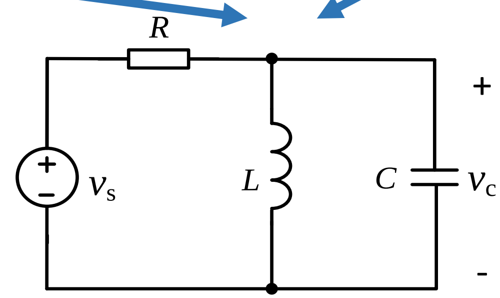
Band-pass

- **Transfer Function and Frequency Response**
- **Filters**
- **Series/Parallel Resonance**

RLC in Series and in Parallel



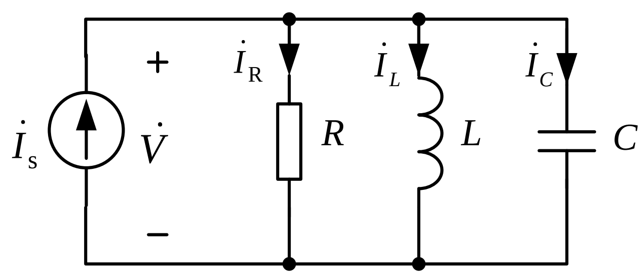




Bandpass

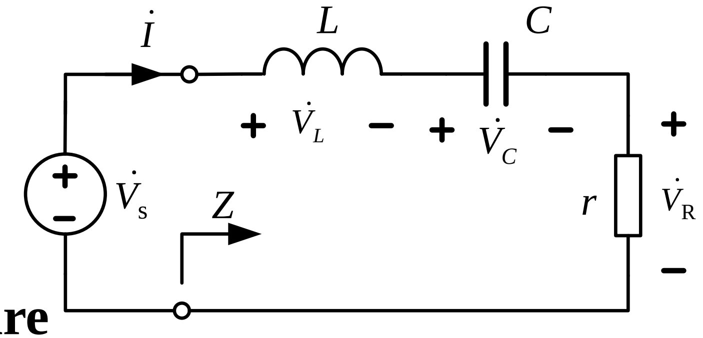
Bandpass

Natural Frequency: $\omega_0 = \dfrac{1}{\sqrt{LC}}$



Frequency Response of RLC in Series

- **Voltage Ratio Transfer**



$$H(\mathrm{j}\omega)=k\frac{\frac{\omega_0}{Q}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\frac{\omega_0}{Q}\mathrm{j}\omega+\omega_0^2}$$

Compare Band-pass

$$H(\mathrm{j}\omega)=\frac{\dot{V}_R}{\dot{V}_s}=\frac{r}{r+\mathrm{j}\omega L+\frac{1}{\mathrm{j}\omega C}}=\frac{\frac{r}{L}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\frac{r}{L}\mathrm{j}\omega+\boxed{\frac{1}{LC}}}$$

$\omega_0{}^2$

$\frac{\omega_0}{Q}$

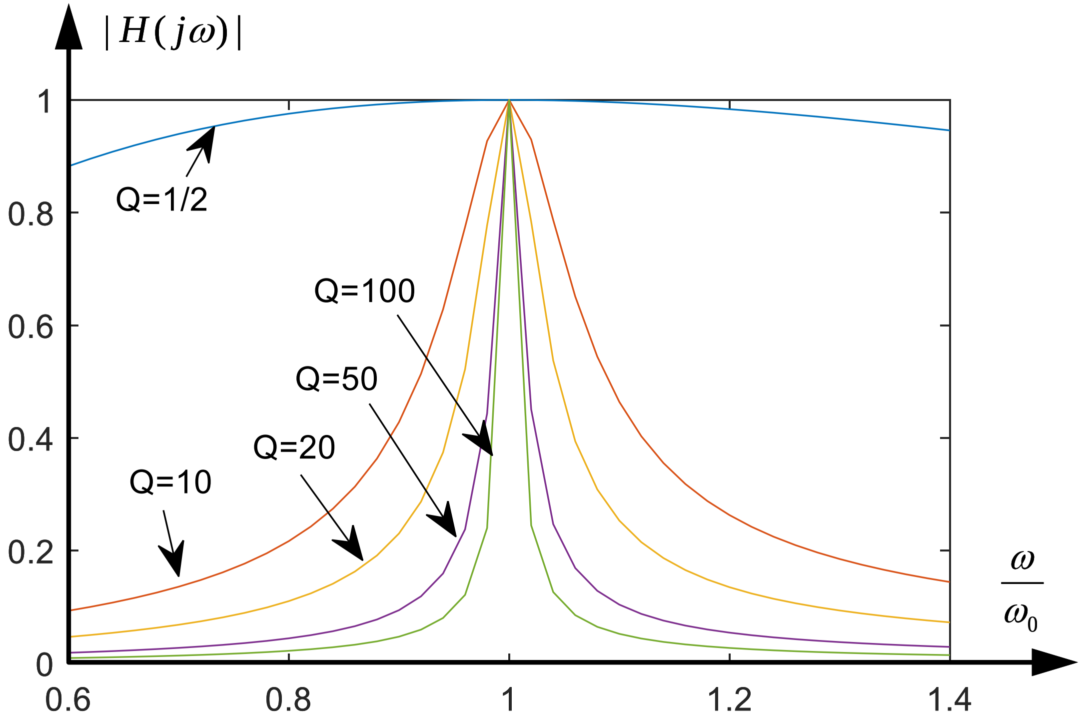
$$B=\omega_2-\omega_1=\frac{\omega_0}{Q}=\frac{r}{L}=\omega_0{}^2Cr$$

'high Q' circuit: $Q>10$, $B\ll\omega_0$, 'narrow band' circuit

$$\omega_1,\omega_2\approx\omega_0\mp\frac{B}{2}$$

The Normalization Amplitude Frequency Characteristics in RLC Series Circuits

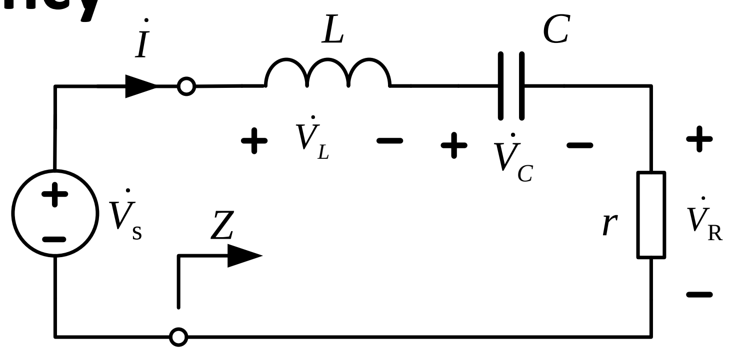
- **The influence of Q in circuits** $B=\omega_2-\omega_1=\dfrac{\omega_0}{Q}$



The higher the value of *Q*, the more selective the circuit is but the smaller the bandwidth.

RLC Series Resonance

- **Resonance Condition and Frequency**



$$Z(\omega)=\frac{\dot{V}_s}{\dot{I}}=r+\mathrm{j}(\omega L-\frac{1}{\omega C})$$

Condition： Im[Z] = 0

Resonance Frequency： $\omega_0=\frac{1}{\sqrt{LC}}$

- **Resonance characteristics**

1. The impedance is purely resistive, thus, Z = R.

The reactance is called characteristic impedance:

$$\rho=\omega_0 L=\frac{1}{\omega_0 C}=\sqrt{\frac{L}{C}}$$

2. The voltage V_S and the current I are in phase, λ_{pf} =cosθ=1.

3. ∣Z(ω)∣= r is minimum.

4. V_L and V_C can be much more than V_S .

Series Resonance Quality Factor

- **Definition of *Q***

$$Q = 2\pi \frac{\text{Peak energy stored in the circuit}}{\text{Energy dissipated by the circuit in one period at resonance}}$$

- ***Q* in RLC series resonance circuits**

$$Q = 2\pi \frac{L I_{\mathrm{m}}^{2}/2}{T_0 r I_{\mathrm{m}}^{2}/2} = \frac{\omega_0 L}{r} = \frac{1}{\omega_0 C r} = \frac{\rho}{r}$$

- ***V*c and *V*$_{\mathrm{L}}$ in RLC series resonance circuits**

$$V_{\mathrm{C}} = V_{\mathrm{L}} = \frac{V_{\mathrm{S}}}{r}\rho = Q V_{\mathrm{S}} \qquad \text{'high } Q\text{'}: \ Q > 10$$

- ***Q* and *B* in RLC series resonance circuits**

$$B = \omega_2 - \omega_1 = \frac{\omega_0}{Q} = \frac{r}{L} = \omega_0^{2} C r$$

Example **In RLC series resonance circuit, *L*=1H, *C* =1μF, find resonance frequency and characteristic impedance. Calculate quality factor, bandwidth and cutoff frequency when *r* =10 Ω and *r* =100Ω.**

solution $\omega_0 = 1/\sqrt{LC} = 10^3 \text{rad/s}$ $\rho = \sqrt{\dfrac{L}{C}} = \sqrt{\dfrac{1}{10^{-6}}} = 1\ \text{k}\Omega$

（1）r =10 Ω

$$Q = \frac{\rho}{r} = \frac{10^3}{10} = 100 \qquad B = \frac{\omega_0}{Q} = \frac{10^3}{100} = 10\ \text{rad/s}$$

$$\omega_1 \approx \omega_0 - \frac{B}{2} = 995\ \text{rad/s} \qquad \omega_2 \approx \omega_0 + \frac{B}{2} = 1005\ \text{rad/s}$$

（2）r =100 Ω

$$Q = \frac{\rho}{r} = \frac{10^3}{100} = 10 \qquad B = \frac{\omega_0}{Q} = \frac{10^3}{10} = 100$$

$$\omega_1 \approx \omega_0 - \frac{B}{2} = 950\ \text{rad/s} \qquad \omega_2 \approx \omega_0 + \frac{B}{2} = 1050\ \text{rad/s}$$

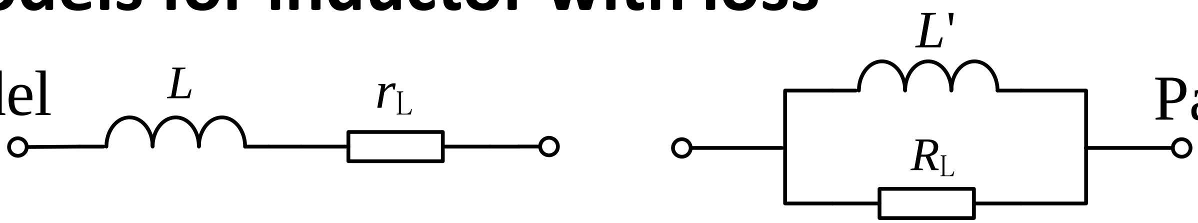
Quality Factor for a real inductor

- **Definition**

$$Q = 2\pi \frac{\text{Peak energy stored in the element}}{\text{Energy dissipated by the element in one period}}$$

- **Two models for inductor with loss**

Series model



Parallel model

(a) (b)

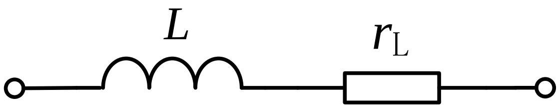
$$Q_L = 2\pi \frac{\frac{1}{2} L I_m^2}{T \cdot \frac{1}{2} r_L I_m^2} = \frac{\omega L}{r_L} \qquad Q_L' = 2\pi \frac{\frac{1}{2} L' \left(\frac{U_m}{\omega L'}\right)^2}{T \frac{1}{2} \frac{U_m^2}{R_L}} = \frac{R_L}{\omega L'}$$

（1）Resistance can be calculated after testing L and Q;

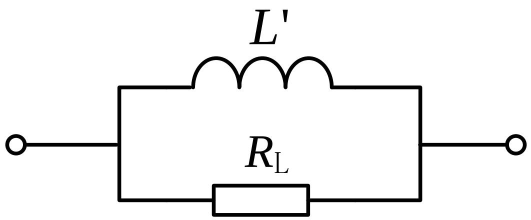
（2）*Q* is related to frequency but keeps constant in a certain range.

Quality Factor for a real inductor

- **Series model and parallel model have the same Q value when Q is very high**



(a)



(b)

$$Q_L = \frac{\omega L}{r_L} \qquad Q_L' = \frac{R_L}{\omega L'}$$

Equivalence for the two models:

$$Y_{\text{series}} = \frac{1}{r_L + j\omega L} = \frac{r_L}{r_L^2 + (\omega L)^2} - j\frac{\omega L}{r_L^2 + (\omega L)^2} \qquad Y_{\text{parallel}} = \frac{1}{R_L} - j\frac{1}{\omega L'}$$

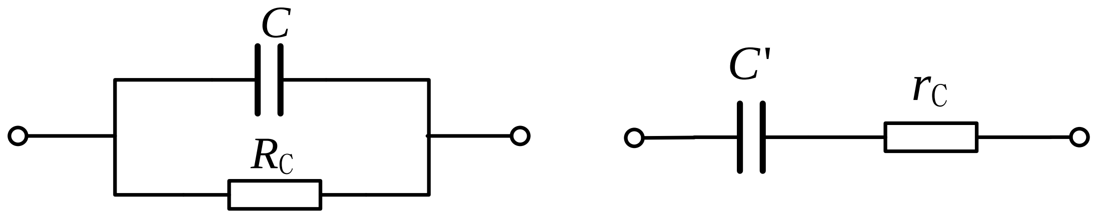
$$R_L = \frac{r_L^2 + (\omega L)^2}{r_L} = (1 + Q_L^2) r_L \approx Q_L^2 r_L = \frac{(\omega L)^2}{r_L}$$

When Q is high

$$\omega L' = \frac{r_L^2 + (\omega L)^2}{\omega L} = (\frac{1}{Q_L^2} + 1)\omega L \approx \omega L$$

Quality Factor for a real capacitor

- **Two models for capacitor with loss**



(a) (b)

Parallel model:

$$Q_C = 2\pi \frac{\frac{1}{2} C V_m^2}{T \cdot \frac{1}{2} V_m^2 / R_C} = \omega C R_C$$

When Q is high $\quad C' = C \qquad r_C = \dfrac{1}{\omega C Q_C}$

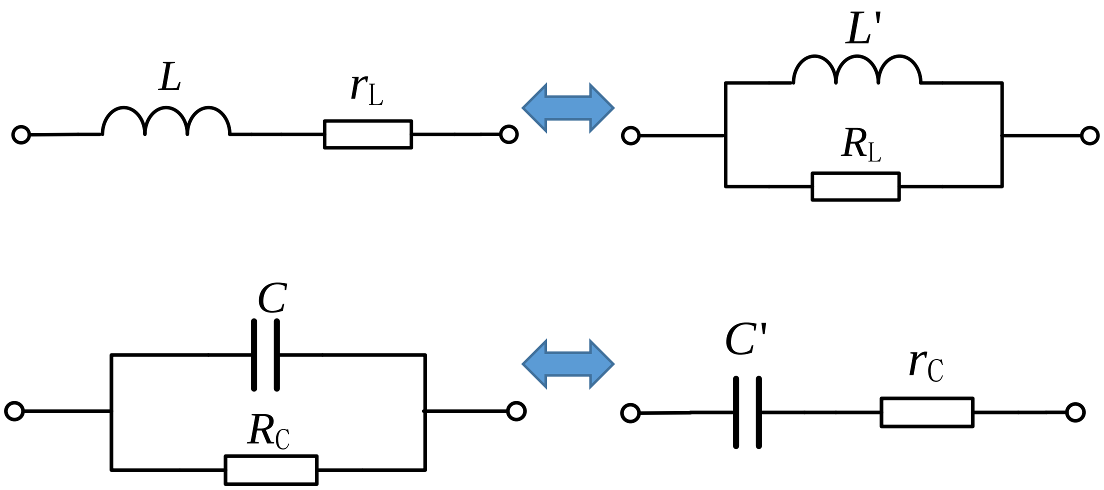
$$R_C = \frac{1}{(\omega C)^2 r_C} = Q_C^2 r_C$$

Summary for Quality Factor of a real reactor

- **Series resistance *r*, parallel resistance *R*, reactance *X***

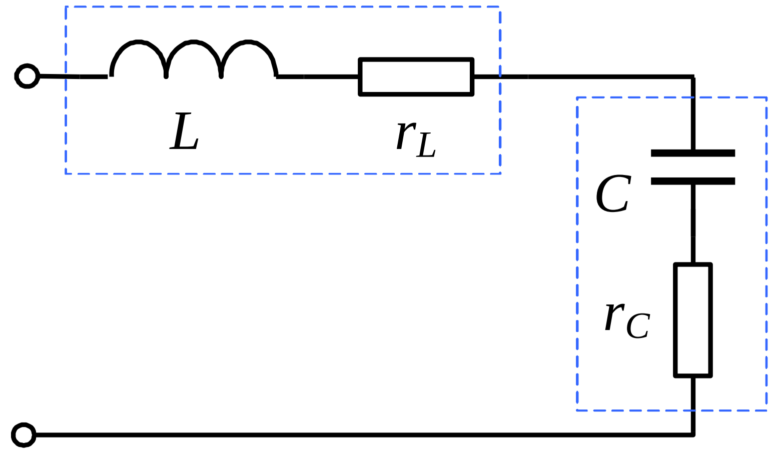
Under high *Q*， *Q* =*Q′*， *L* =*L′*， *C* =*C′*， and

$$r = \frac{|X|}{Q} \qquad R = Q|X| \qquad R = \frac{X^2}{r} = Q^2 r$$



Q Value in Series Resonance

Considering real L and C in LC circuit



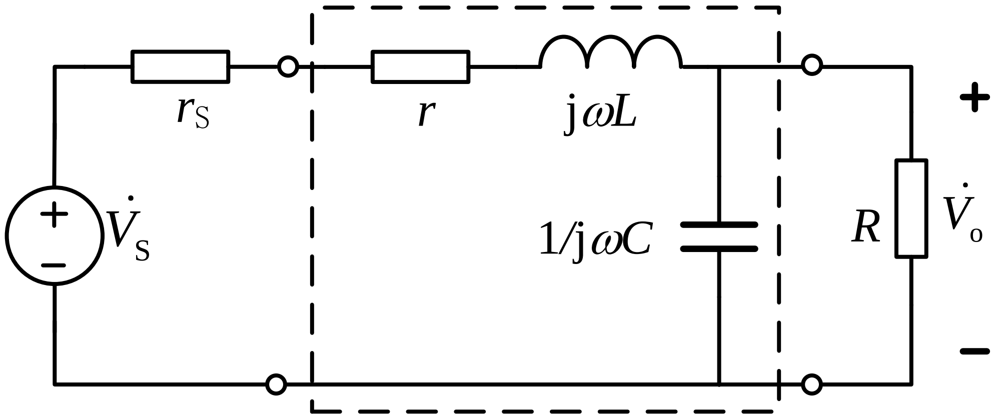
$$r_{\mathrm{L}} = \frac{X_{\mathrm{L}}}{Q_{\mathrm{L}}} = \frac{\rho}{Q_{\mathrm{L}}}$$

$$r_{\mathrm{C}} = \frac{|X_{\mathrm{C}}|}{Q_{\mathrm{C}}} = \frac{\rho}{Q_{\mathrm{C}}}$$

***Q* for the circuit:** $$Q = \frac{\rho}{r} = \frac{\rho}{r_{\mathrm{L}} + r_{\mathrm{C}}} = \frac{\rho}{\frac{\rho}{Q_{\mathrm{L}}} + \frac{\rho}{Q_{\mathrm{C}}}} = \frac{Q_{\mathrm{L}} Q_{\mathrm{C}}}{Q_{\mathrm{L}} + Q_{\mathrm{C}}}$$

Normally, $Q_{\mathrm{C}} >> Q_{\mathrm{L}}$ **so** $Q \approx Q_{\mathrm{L}}$

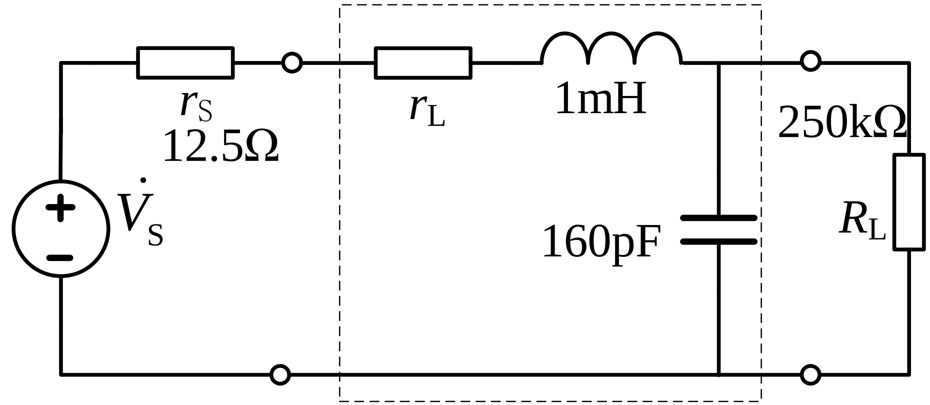
Effect on Q From Source and Load



- **Internal resistance in source and load increases the loss in circuit;**
- **Analysis method: take all resistors into equivalent series resistance in LC circuit.**

Example **Given** Q_L=200, Q_C>>Q_L, V_s=10mV(rms), the circuit works in resonance. Find Q and V_{C0} with or without load.

solution



(1)Consider only the circuit in the block

$$Q \approx Q_L = 200$$

$$\omega_0 = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10^{-3} \times 160 \times 10^{-12}}} = 2.5 \times 10^6 \text{ rad/s}$$

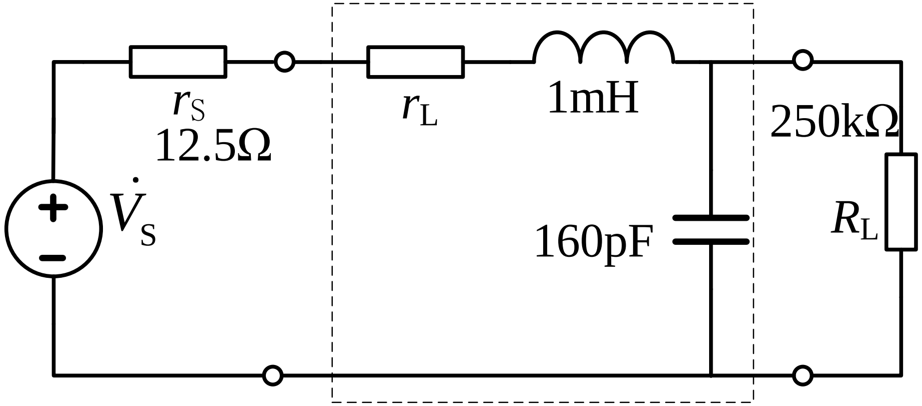
$$\rho = \sqrt{L/C} = \sqrt{10^{-3}/160 \times 10^{-12}} = 2.5 \times 10^3 \ \Omega$$

$$r_L = \frac{\rho}{Q_L} = \frac{\sqrt{L/C}}{Q_L} = \frac{\sqrt{10^{-3}/160 \times 10^{-12}}}{200} = \frac{2.5 \times 10^3}{200} = 12.5 \ \Omega$$

(2) Consider $R_L \to \infty$

$$Q_1 = \frac{\rho}{r_S + r_L} = \frac{2.5\times 10^3}{12.5+12.5} = 100$$

$$V_{C01} = Q_1 V_S = 1\ \text{V}$$



(3) Consider R_L

Applying parallel model for Q_C:

$$Q_C = \omega C R_C = \frac{R_C}{|X_C|} = \frac{250\text{k}\Omega}{2.5\text{k}\Omega} = 100 \quad \to \text{high Q}$$

Convert parallel model to series model:

$$r_C = \frac{X_C{}^2}{R} = \frac{\rho^2}{R} = \frac{(2.5\times 10^3)^2}{250\times 10^3} = 25\ \Omega$$

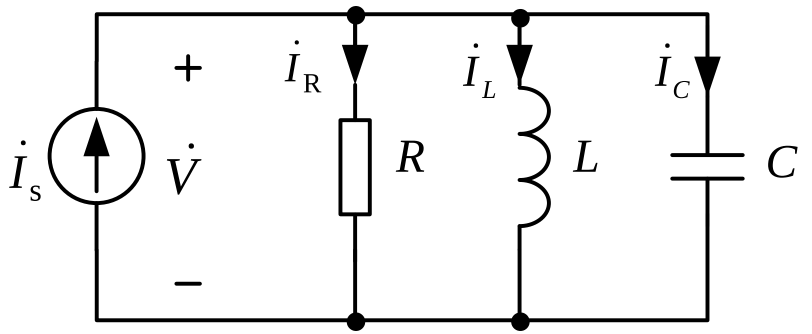
Quality factor as a whole in the circuit:

$$Q_2 = \frac{\rho}{r_S + r_L + r_C} = \frac{2.5\times 10^3}{12.5+12.5+25} = 50 \qquad V_{C02} = Q_2 V_S = 0.5\ \text{V}$$

RLC Parallel Resonance

- **Resonance Condition and Frequency**

$$Y_{\text{in}}(\omega)=\frac{1}{R}+\text{j}(\omega C-\frac{1}{\omega L})$$



Condition： Im[Y] = 0

Resonance Frequency： $\omega_0=\dfrac{1}{\sqrt{LC}}$

- **Resonance characteristics**

1. The impedance is purely resistive, thus, Z = R.

The characteristic impedance: $\rho=\omega_0 L=\dfrac{1}{\omega_0 C}=\sqrt{\dfrac{L}{C}}$

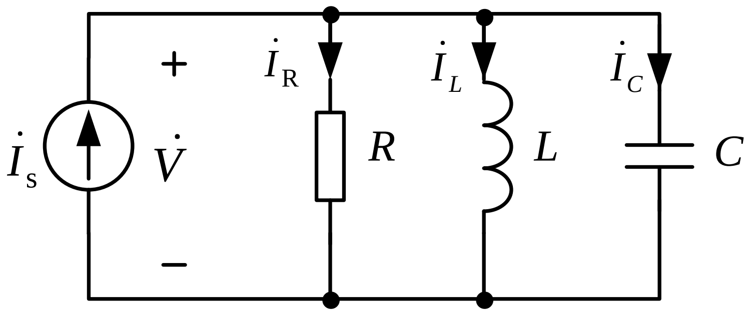
2. The voltage V and the current I_s are in phase, λ_{pf}=cosθ=1.

3. | $Y(\omega)$ | =1/R is minimum， $V\rightarrow$ max.

4. I_L and I_C can be much more than I_S .

RLC Parallel Resonance

- **Quality factor in parallel resonance:**



$$Q = 2\pi \frac{\frac{1}{2} C V_m^2}{T \cdot \frac{1}{2} V_m^2 / R} = \boxed{\omega_0 C R = \frac{R}{\omega_0 L} = \frac{R}{\rho}}$$

- **Current in parallel resonance :**

$$I_C = I_L = \frac{I_s R}{\omega_0 L} = I_s \omega_0 C R = Q I_s$$

$$I_R = I_s$$

RLC Parallel Resonance

- **Input impedance:**

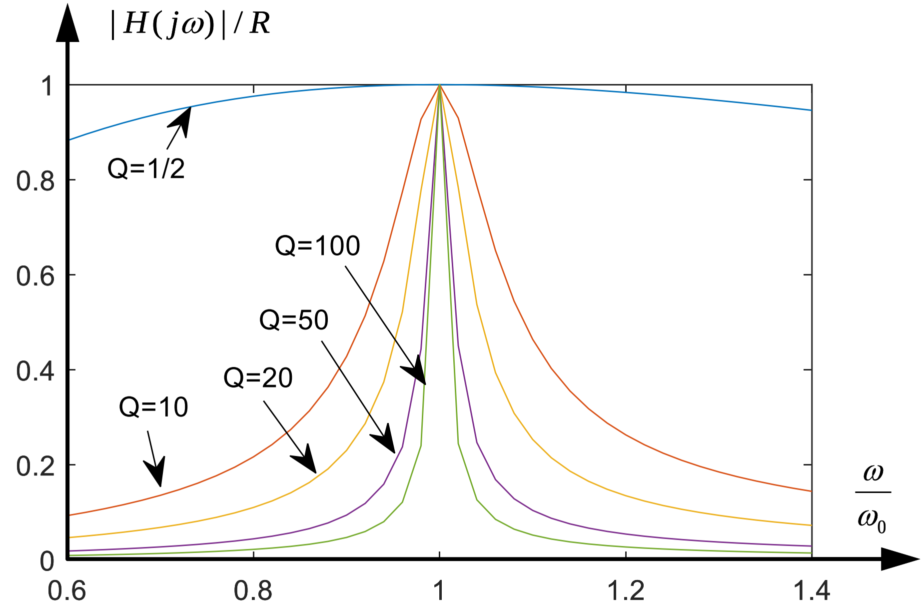
$$H(\mathrm{j}\omega)=\frac{\dot{V}}{\dot{I}_{\mathrm{s}}}=Z(\mathrm{j}\omega)=\frac{1}{\frac{1}{R}+\mathrm{j}\omega C+\frac{1}{\mathrm{j}\omega L}}=\frac{\frac{1}{C}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\frac{1}{RC}\mathrm{j}\omega+\frac{1}{LC}}$$

$$\frac{\omega_0}{Q}$$

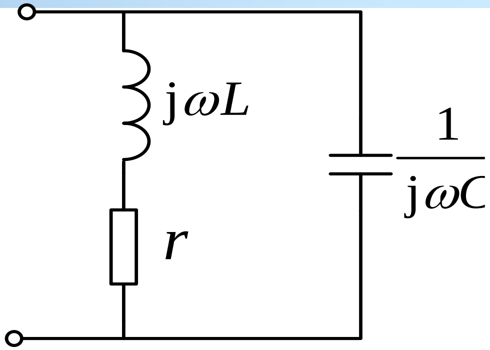
- **Bandpass characteristics:**

$$\omega_0=\frac{1}{\sqrt{LC}} \qquad Q=R\omega_0 C=R/(\omega_0 L)=R/\rho=\omega_0/B \qquad k=R$$

- **Frequency selection and *Q*:**



Practical Parallel Resonance



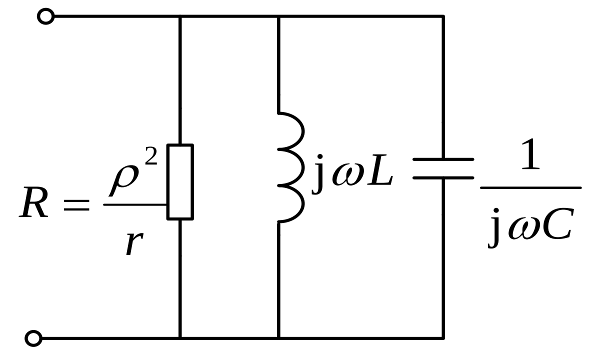
$$Y=\mathrm{j}\omega C+\frac{1}{r+\mathrm{j}\omega L}=\mathrm{j}\omega C+\frac{r-\mathrm{j}\omega L}{r^2+(\omega L)^2}$$

Set $$\mathrm{Im}[Y]=\omega C-\frac{\omega L}{r^2+(\omega L)^2}=0$$

$$R=\frac{X^2}{r}=\frac{\rho^2}{r}=\frac{L}{Cr}$$

Get parallel resonance frequency:

$$\omega_\mathrm{p}=\frac{1}{\sqrt{LC}}\sqrt{1-\frac{r^2}{\rho^2}}=\omega_0\sqrt{1-\frac{1}{Q_\mathrm{L}^{\ 2}}}$$

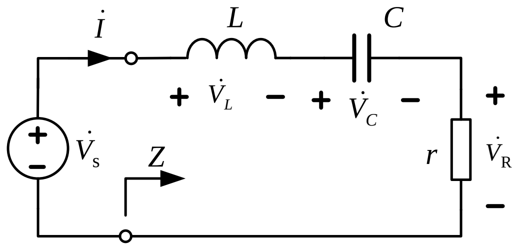


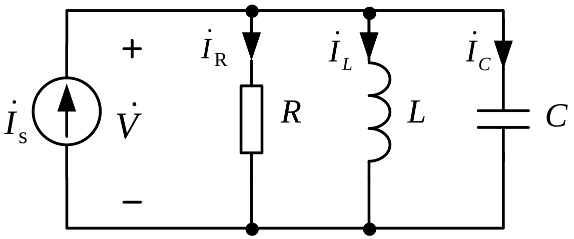
When Q is high, $\omega_\mathrm{p}\approx\omega_0$

$$Q=\frac{R}{\rho}=\frac{\rho}{r}=\frac{\sqrt{L/C}}{r}$$

For high Q inductor, $r \ll \rho$

Recap series and parallel resonance





$$H(\mathrm{j}\omega)=\frac{\dot{V}_R}{\dot{V}_s}=\frac{\frac{r}{L}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\frac{r}{L}\mathrm{j}\omega+\frac{1}{LC}}$$

$$H(\mathrm{j}\omega)=\frac{\dot{V}}{\dot{I}_s}=\frac{\frac{1}{C}\mathrm{j}\omega}{(\mathrm{j}\omega)^2+\frac{1}{RC}\mathrm{j}\omega+\frac{1}{LC}}$$

$$\omega_0=\frac{1}{\sqrt{LC}} \qquad B=\frac{r}{L}$$

$$\omega_0=\frac{1}{\sqrt{LC}} \qquad B=\frac{1}{RC}$$

$$\rho=\omega_0 L=\frac{1}{\omega_0 C}=\sqrt{\frac{L}{C}}$$

$$\rho=\omega_0 L=\frac{1}{\omega_0 C}=\sqrt{\frac{L}{C}}$$

$$Q=\frac{\rho}{r}=\frac{\sqrt{L/C}}{r}=\frac{\omega_0}{B}$$

$$Q=\frac{R}{\rho}=\frac{R}{\sqrt{L/C}}=\frac{\omega_0}{B}$$

$$Q>10,\ B<<\omega_0, \qquad \omega_1,\omega_2\approx\omega_0\mp\frac{B}{2}$$

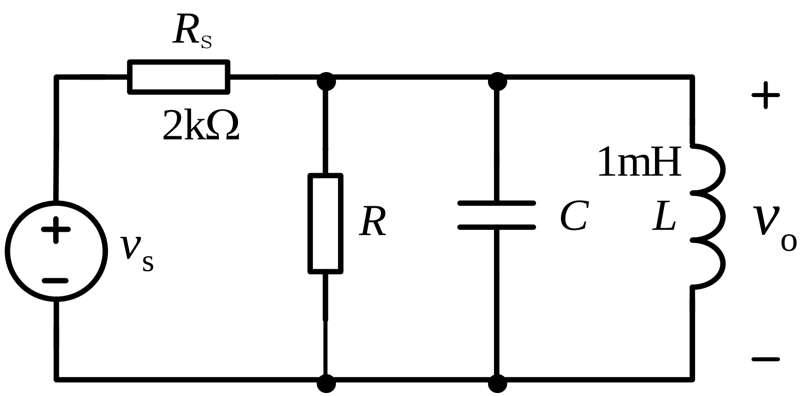
$$\omega_1,\omega_2\approx\omega_0\mp\frac{B}{2}$$

$$V_C=V_L=\frac{V_S}{r}\rho=QV_S$$

$$I_C=I_L=\frac{I_s R}{\rho}=QI_s$$

Example **Given ω_0=10^5rad/s. If required bandwidth is not more than B=10^4rad/s, find C and minimum R.**

solution **From:** $$\omega_0 = \frac{1}{\sqrt{LC}}$$



Get $$C = \frac{1}{\omega_0^2 L} = 100\,\text{nF}$$

Total parallel resistance in the circuit is

$$R_s // R = \frac{R_s \cdot R}{R_s + R}$$

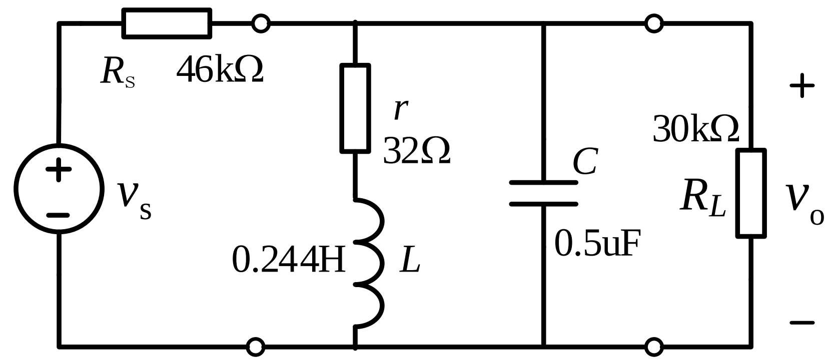
Since bandwidth is not more than B=10^4rad/s, Q is not less than

$$Q = \frac{\omega_0}{B} = 10 \quad \Rightarrow \quad R_s // R = Q\omega_0 L = 1000$$

$$R \geq 2\,\text{k}\Omega$$

Example **Find the resonant frequency, no-load Q , on load Q and B.**

solution (1) Just L(r)C:



$$Q=\frac{\sqrt{L/C}}{r}=\frac{\sqrt{0.244/0.5\times10^{-6}}}{32}=21.8$$

Q>10， high Q

$$f_0=\frac{1}{2\pi\sqrt{LC}}=\frac{1}{6.28\sqrt{0.244\times0.5\times10^{-6}}}=456\text{ Hz}$$

$$B_f=\frac{f_0}{Q_1}=\frac{456}{21.8}=20.9\text{ Hz}$$

(2) No load→ Convert resistor in series with inductor to that in parallel.

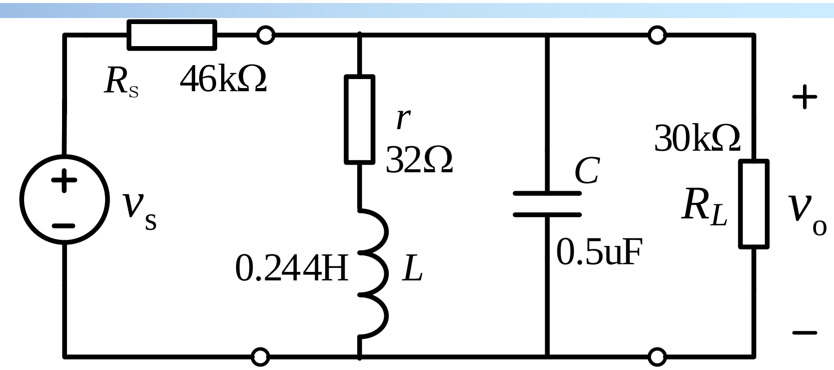
$$R=\frac{\rho^2}{r}=\frac{L}{Cr}=\frac{0.244}{0.5\times10^{-6}\times32}=15.3\text{ k}\Omega$$

$$Q_1=\frac{R_{//}}{\rho}=\frac{1}{\frac{1}{R}+\frac{1}{R_S}}\frac{1}{\sqrt{\frac{L}{C}}}=\frac{1}{(\frac{1}{15.3\text{k}}+\frac{1}{46\text{k}})699}=16.4$$

$$B_f=\frac{f_0}{Q_1}=\frac{456}{16.4}=27.8\text{ Hz}$$

Example **Find the resonant frequency, no-load Q , on load Q and B.**

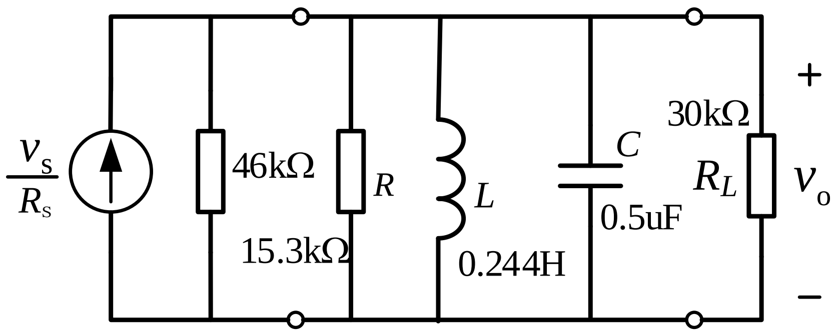
solution



(3) On load

$$Q_2 = \frac{R_{//}}{\rho} = \frac{1}{\frac{1}{R}+\frac{1}{R_S}+\frac{1}{R_L}}\frac{1}{\sqrt{\frac{L}{C}}} = \frac{1}{\left(\frac{1}{15.3k}+\frac{1}{46k}+\frac{1}{30k}\right)699} = 11.8$$

$$B_f = \frac{f_0}{Q'} = \frac{456}{11.8} = 38.6\,\text{Hz}$$



Homework

- **15, 17, 19, 20, 21**